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(54) **SECURE L1/L2 CENTRIC INTER-CELL
MOBILITY EXECUTION**

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H04W 36/08 (2009.01)

(52) **U.S. Cl.**
CPC **H04W 36/0061** (2013.01); **H04W 36/08**
(2013.01)

(58) **Field of Classification Search**

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36/08; H04W 36/0077; H04W 36/00;
H04W 76/11; H04W 36/0085
See application file for complete search history.

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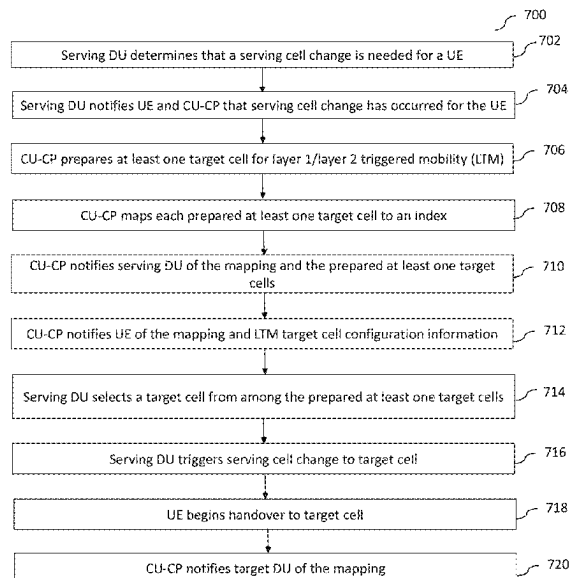
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(57) **ABSTRACT**

In general, the current subject matter relates to secure layer 1/layer 2 (L1/L2) centric inter-cell mobility execution. In some implementations, secure L1/L2 centric inter-cell mobility execution can include mapping a physical cell identifier (PCI) of at least one L1/L2 triggered mobility (LTM) target cell of at least one target distributed unit (DU) of a base station that is a candidate for handover (HO) for a user equipment (UE) to an index, and securely transmitting the mapping from a centralized unit control plane (CU-CP) of the base station to the UE and to a serving DU of the base station that is currently serving the UE for at least one service.

19 Claims, 16 Drawing Sheets



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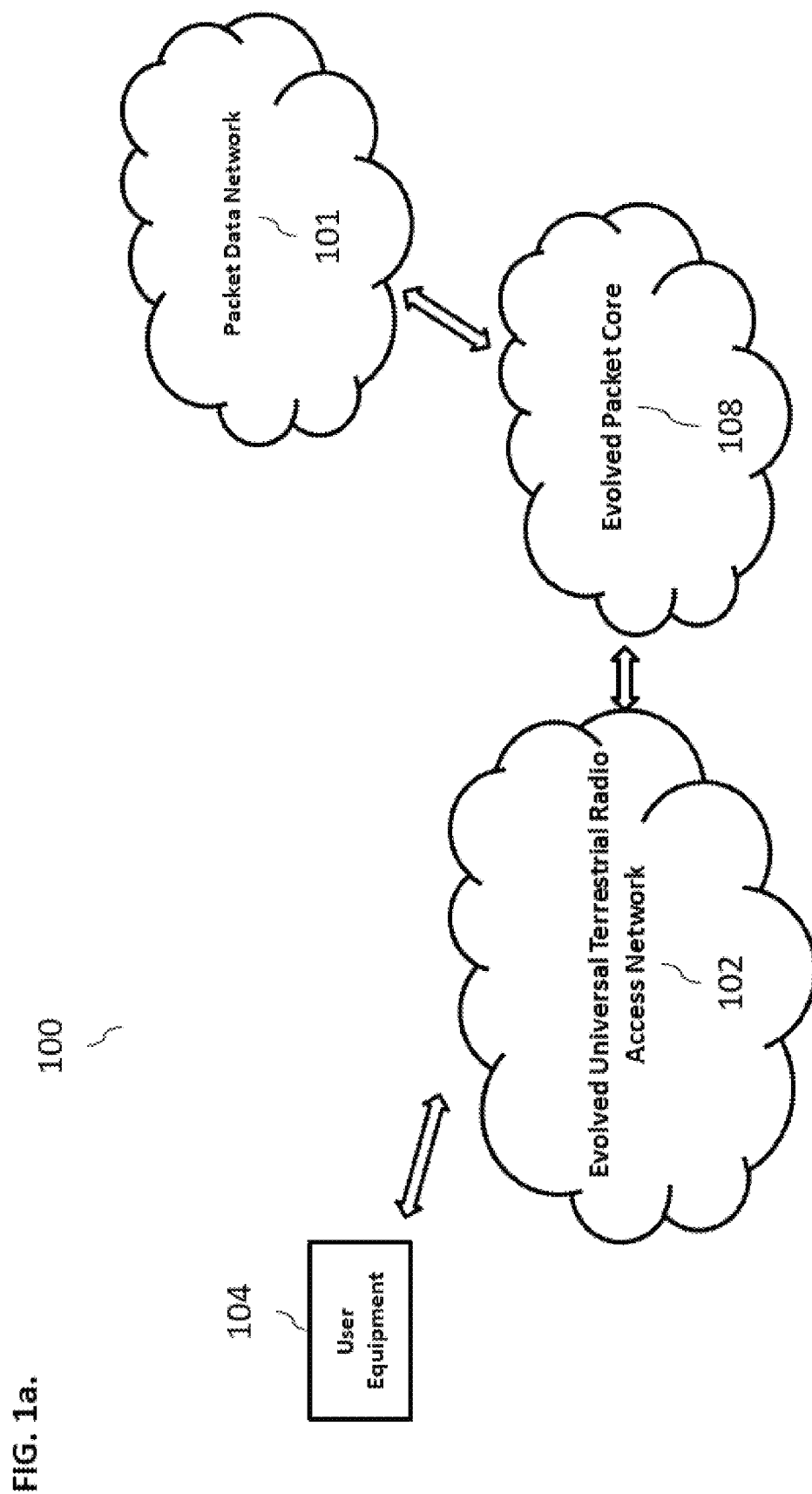


FIG. 1b.

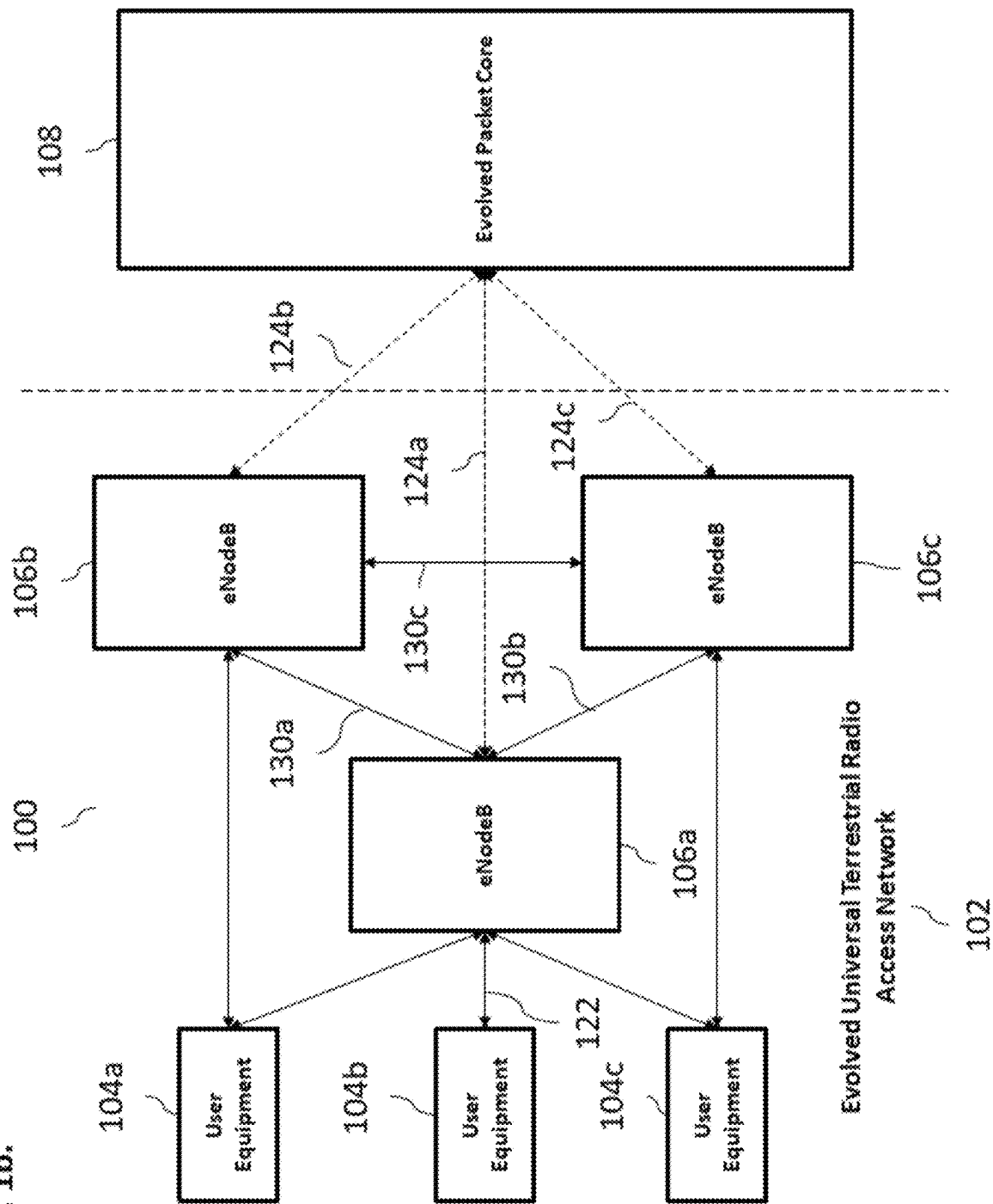


FIG. 1c.

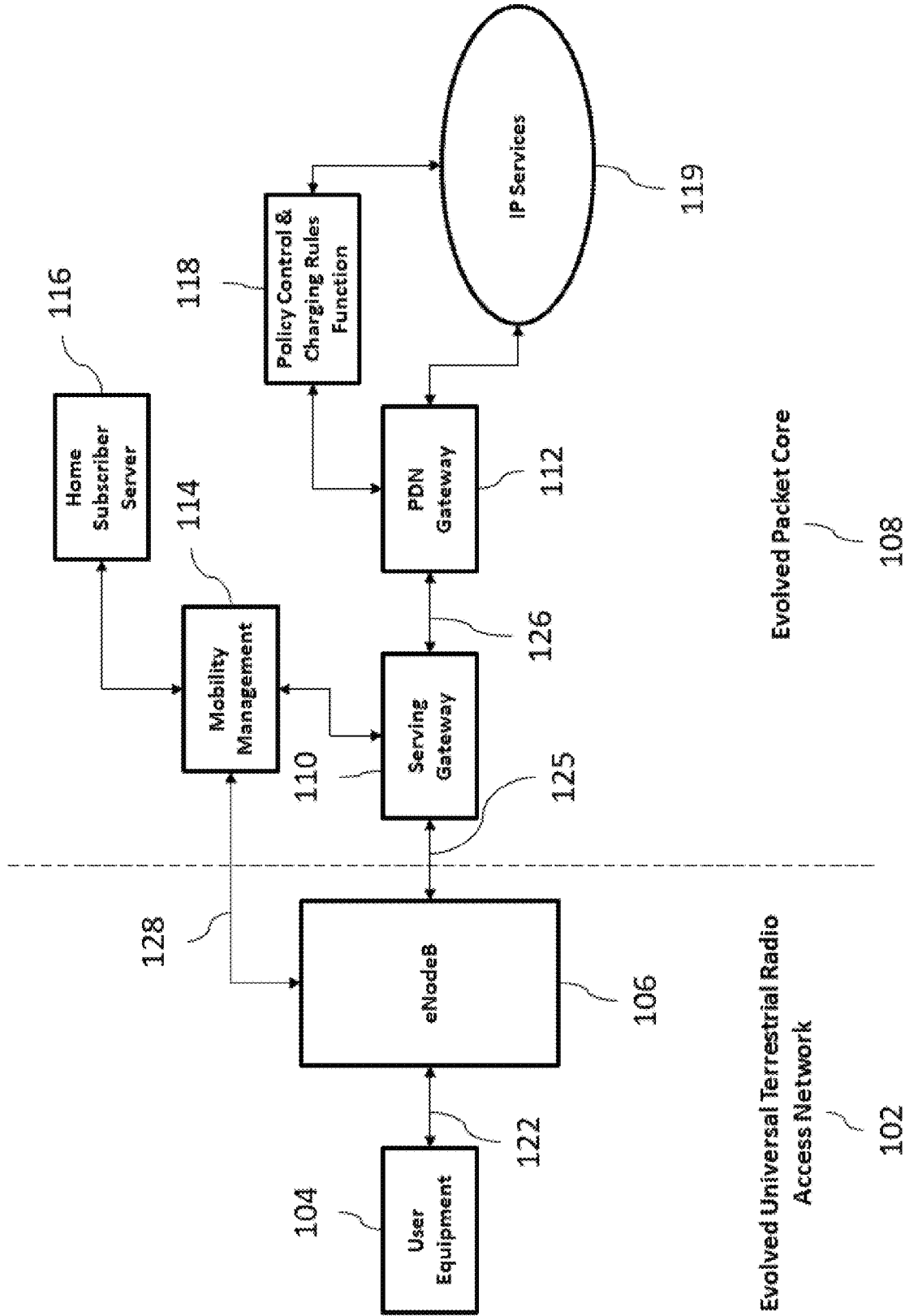
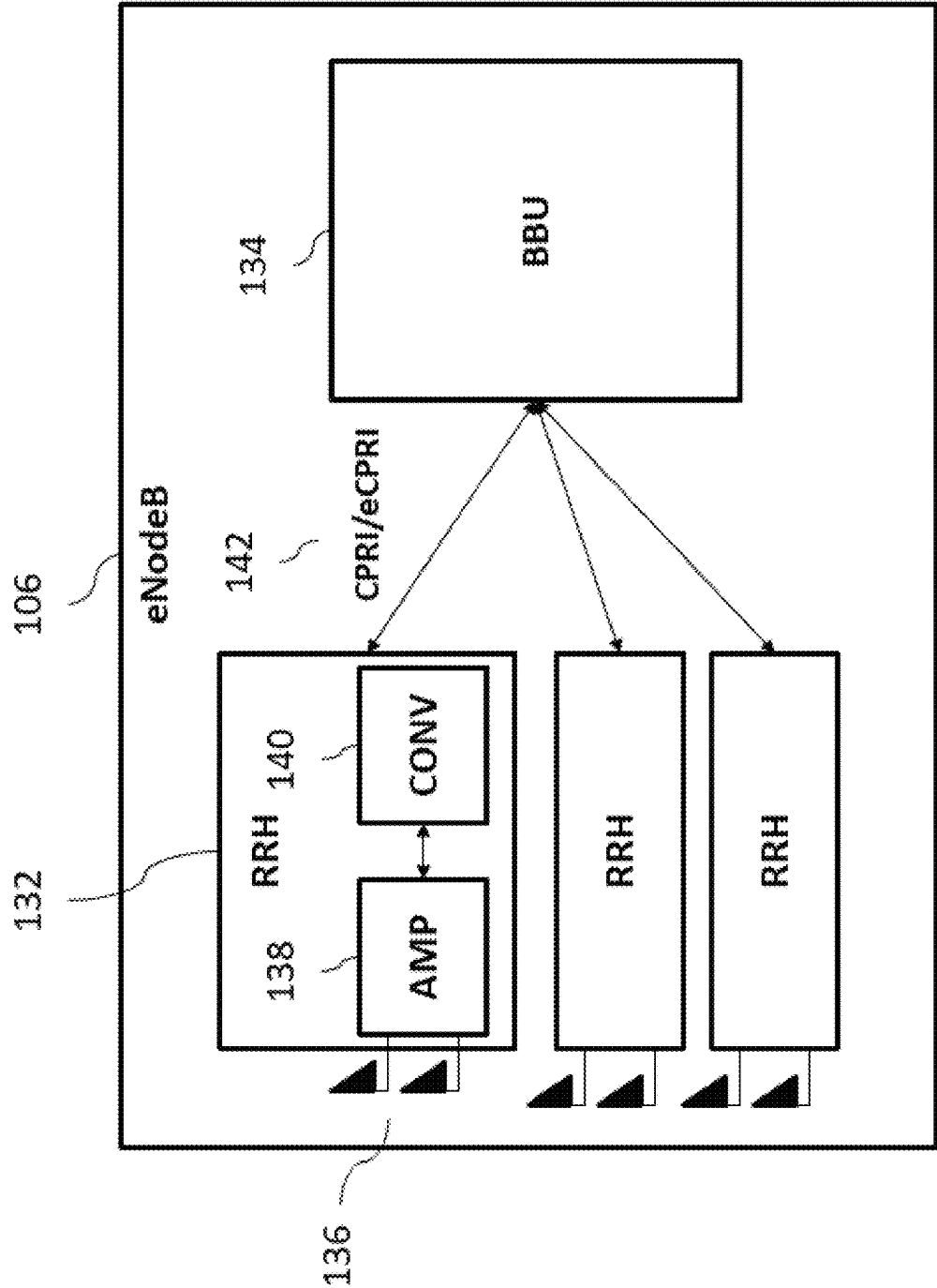


FIG. 1d.



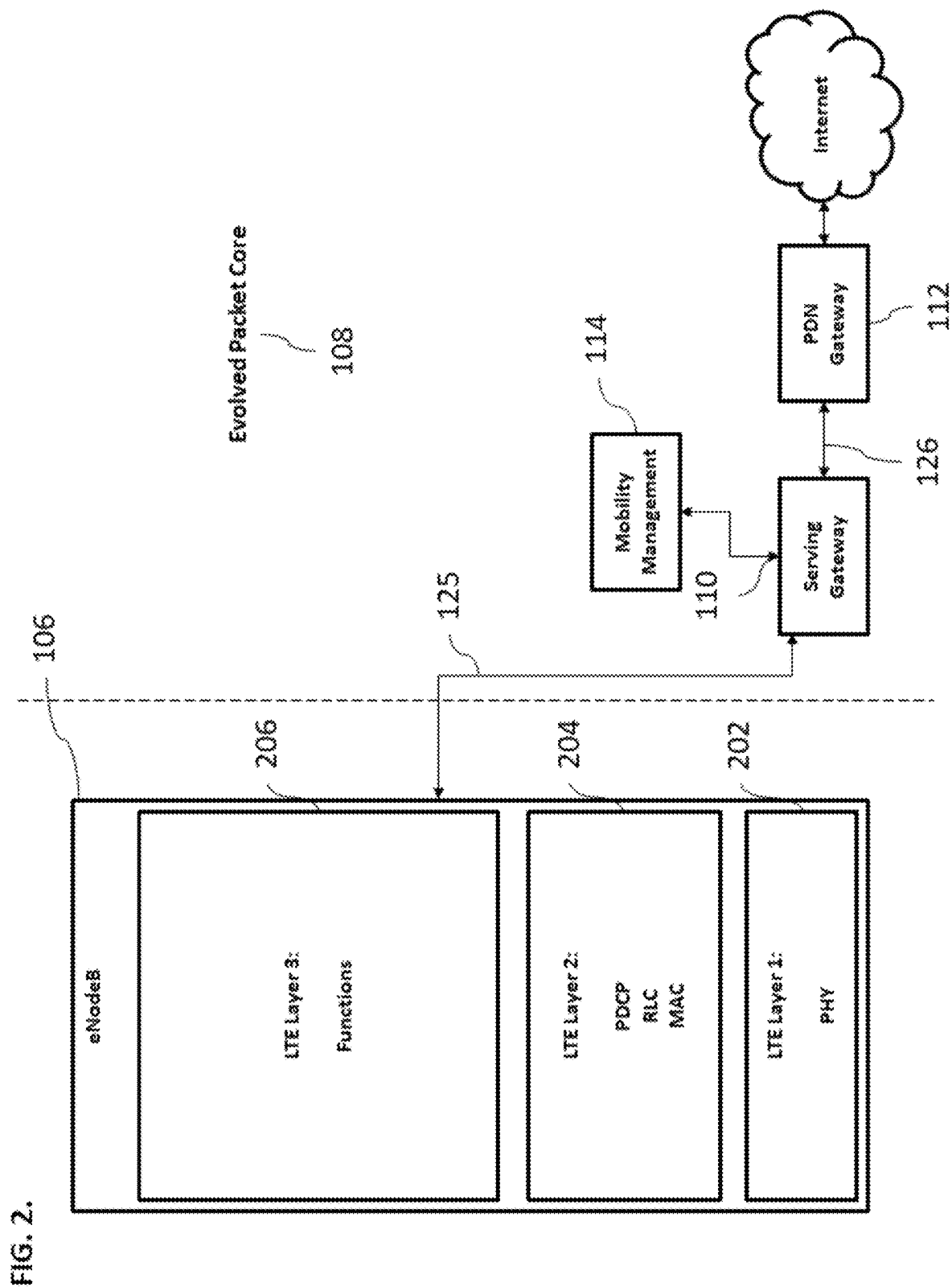


FIG. 3.

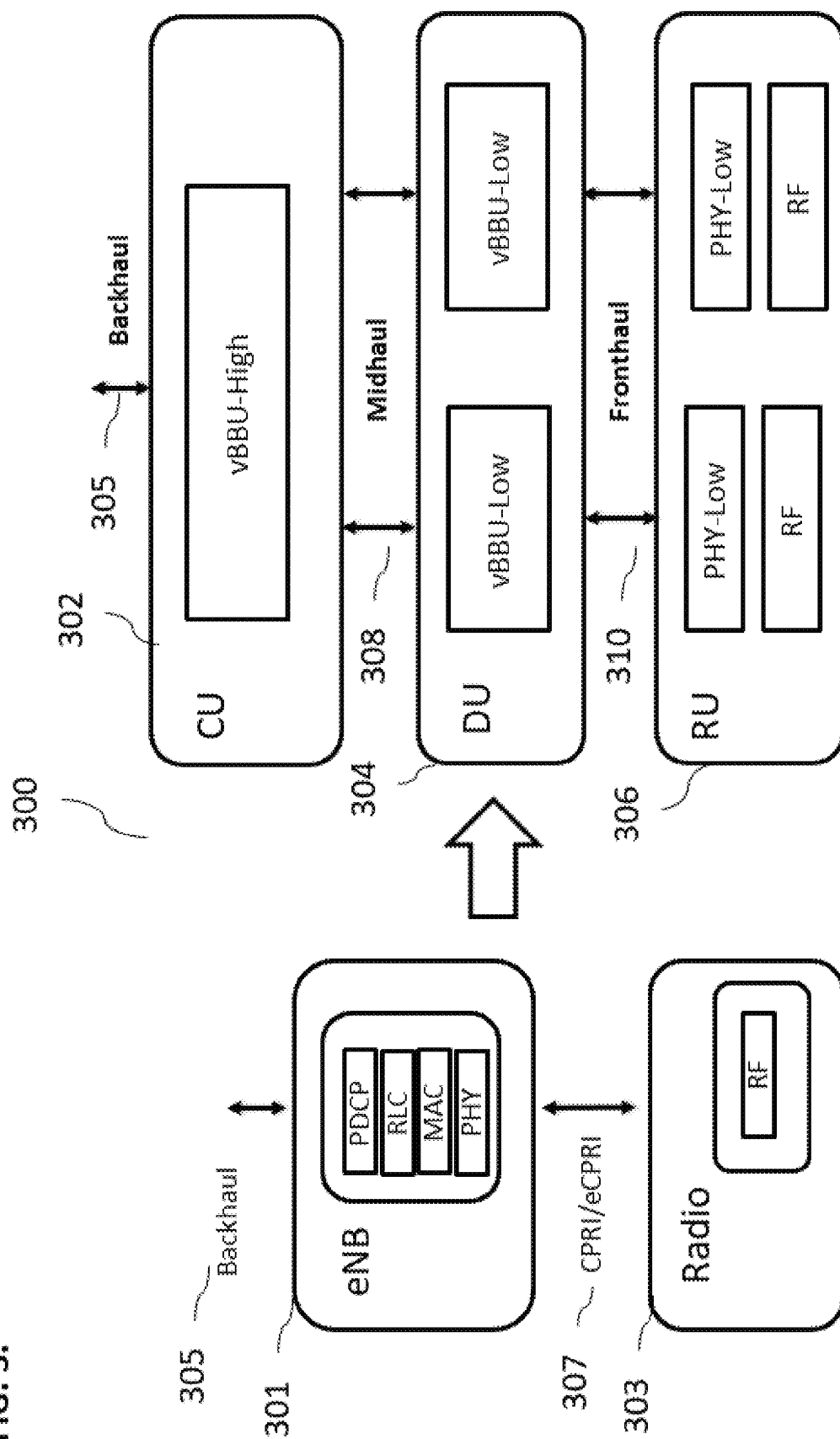


FIG. 4.

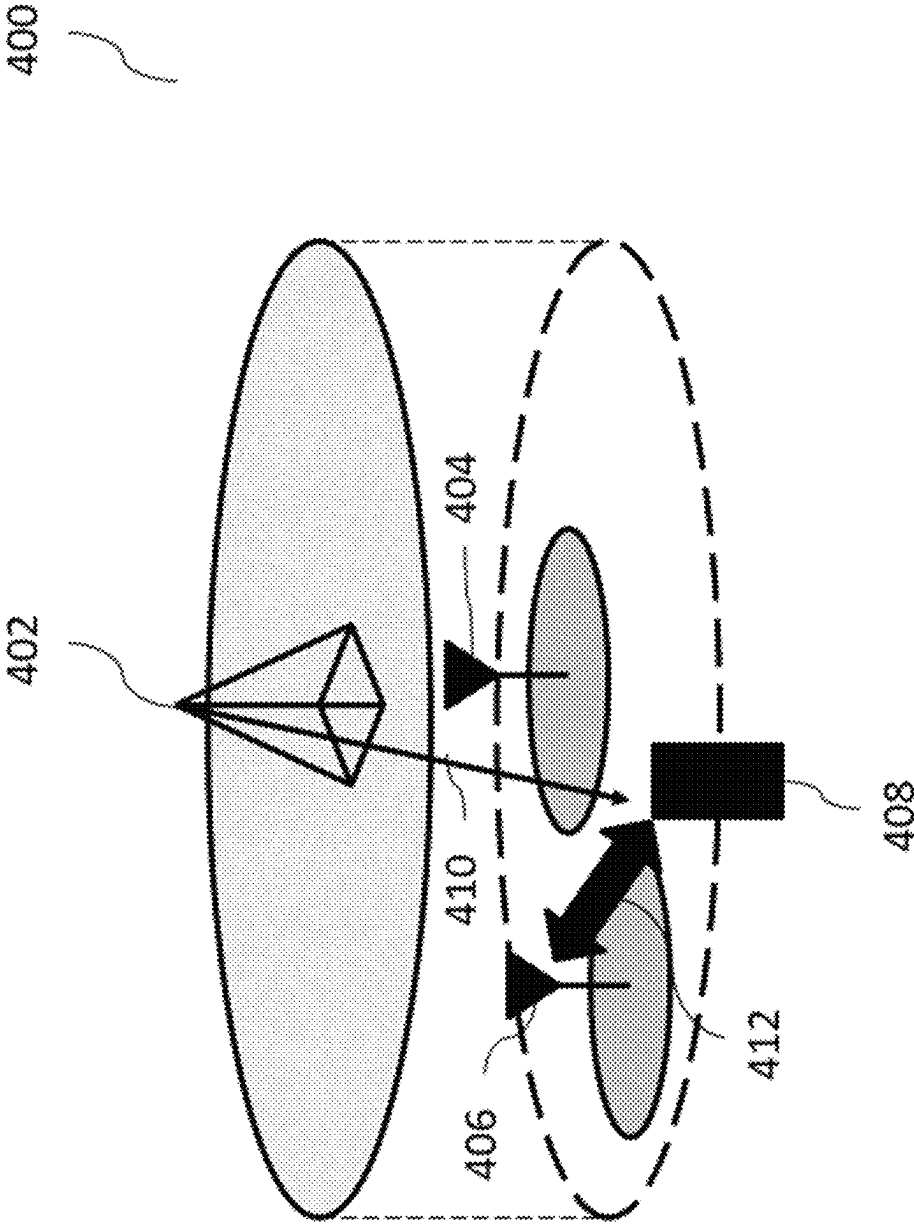


FIG. 5a.

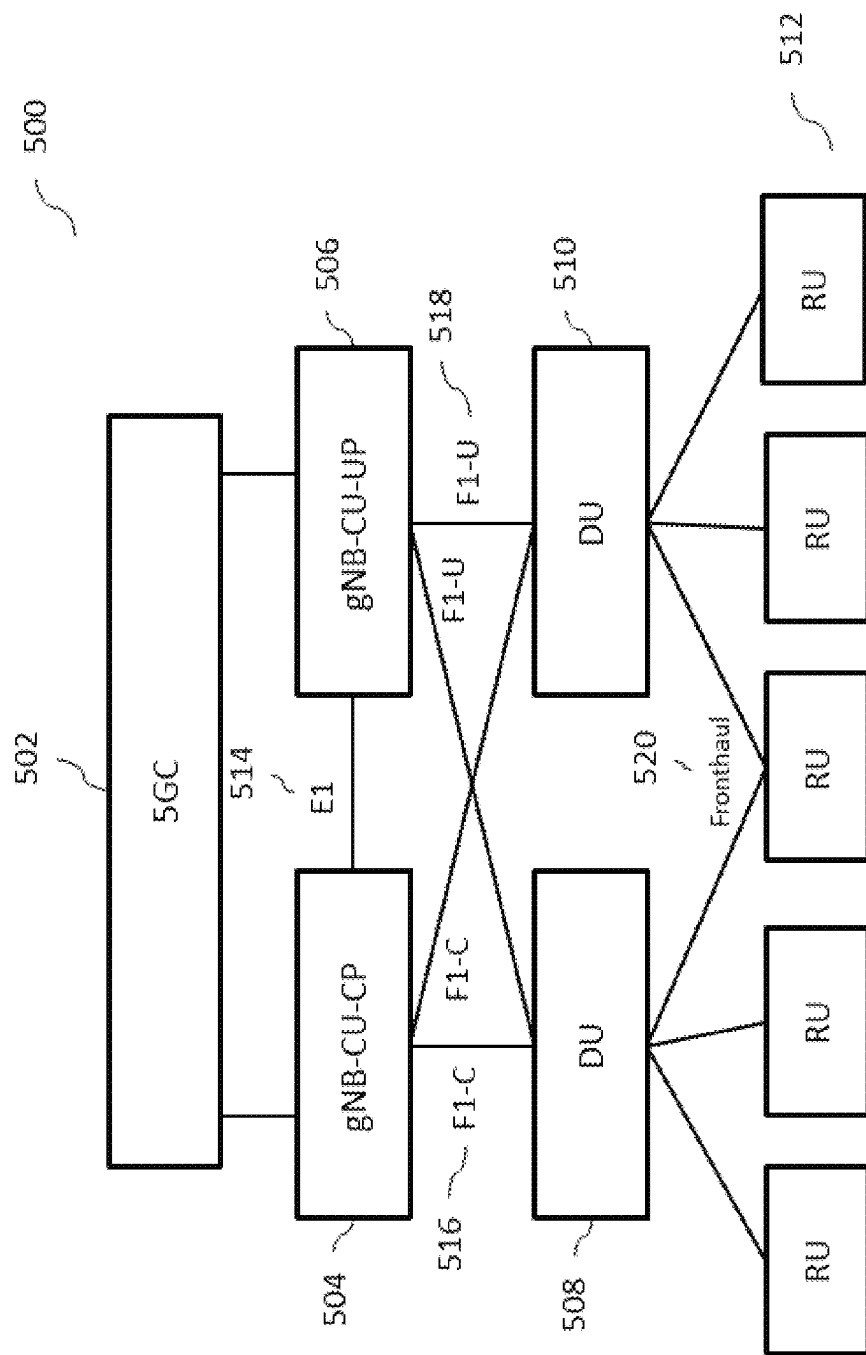


FIG. 5b.

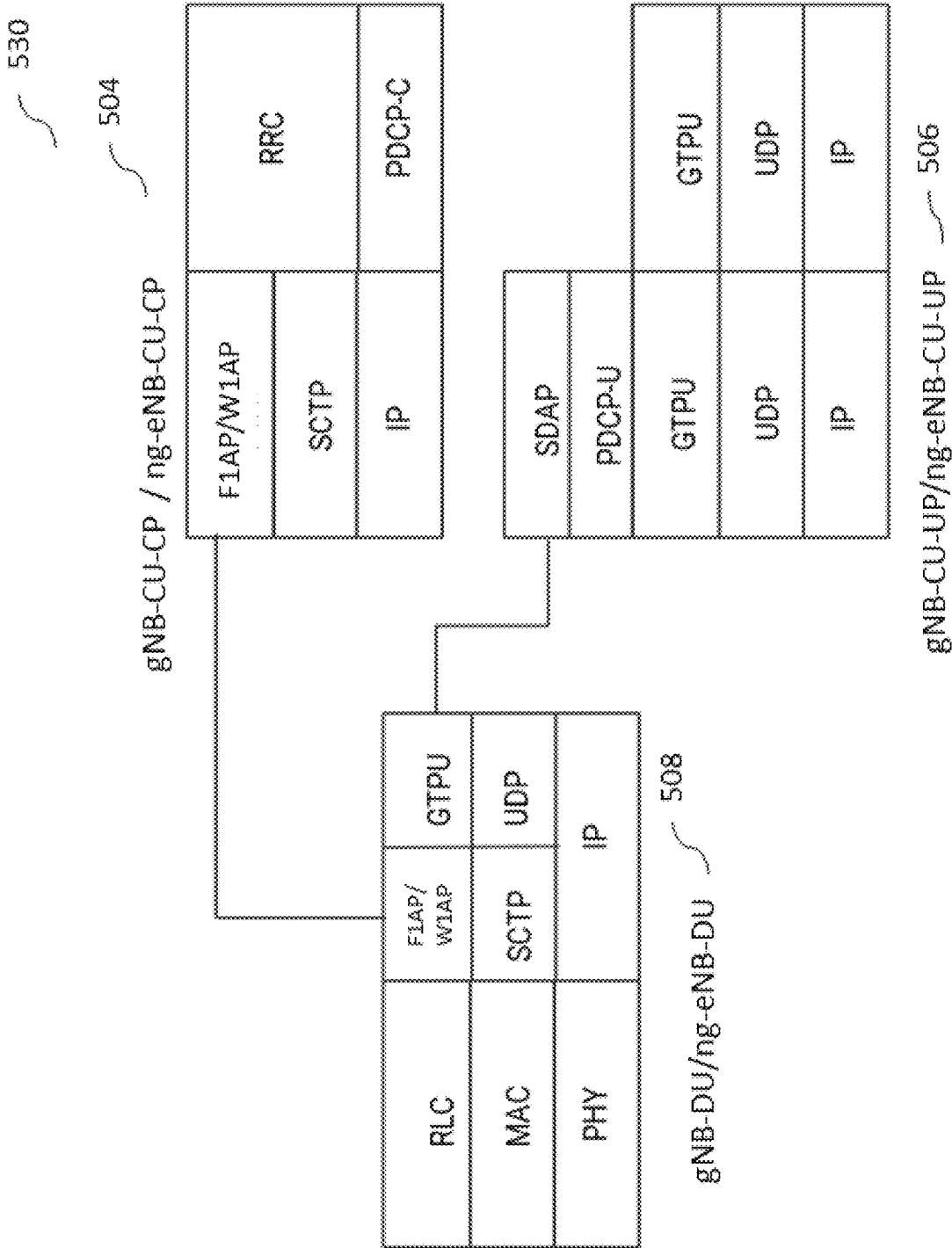


FIG. 5c.

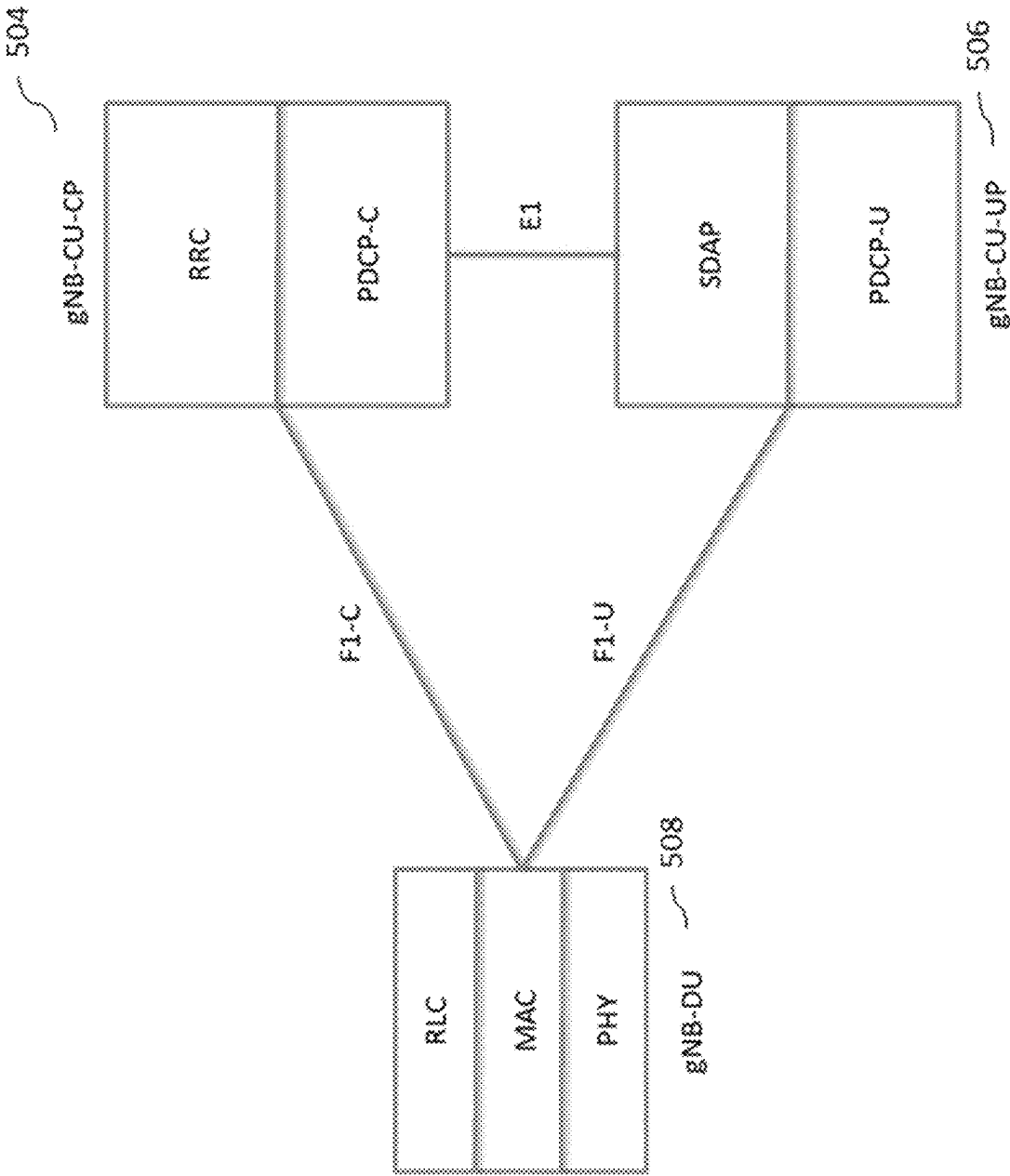


FIG. 6a.

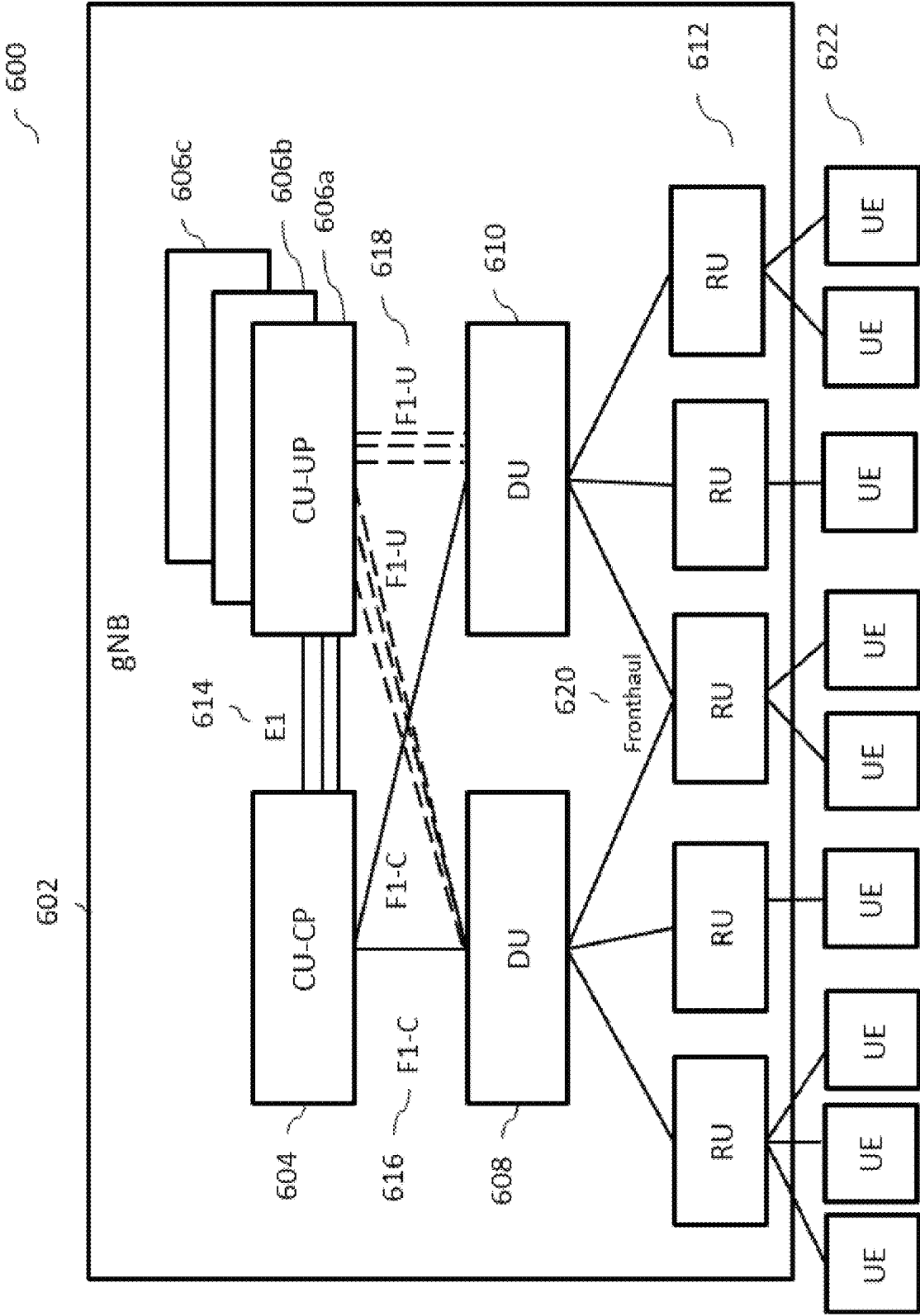


FIG. 6b.

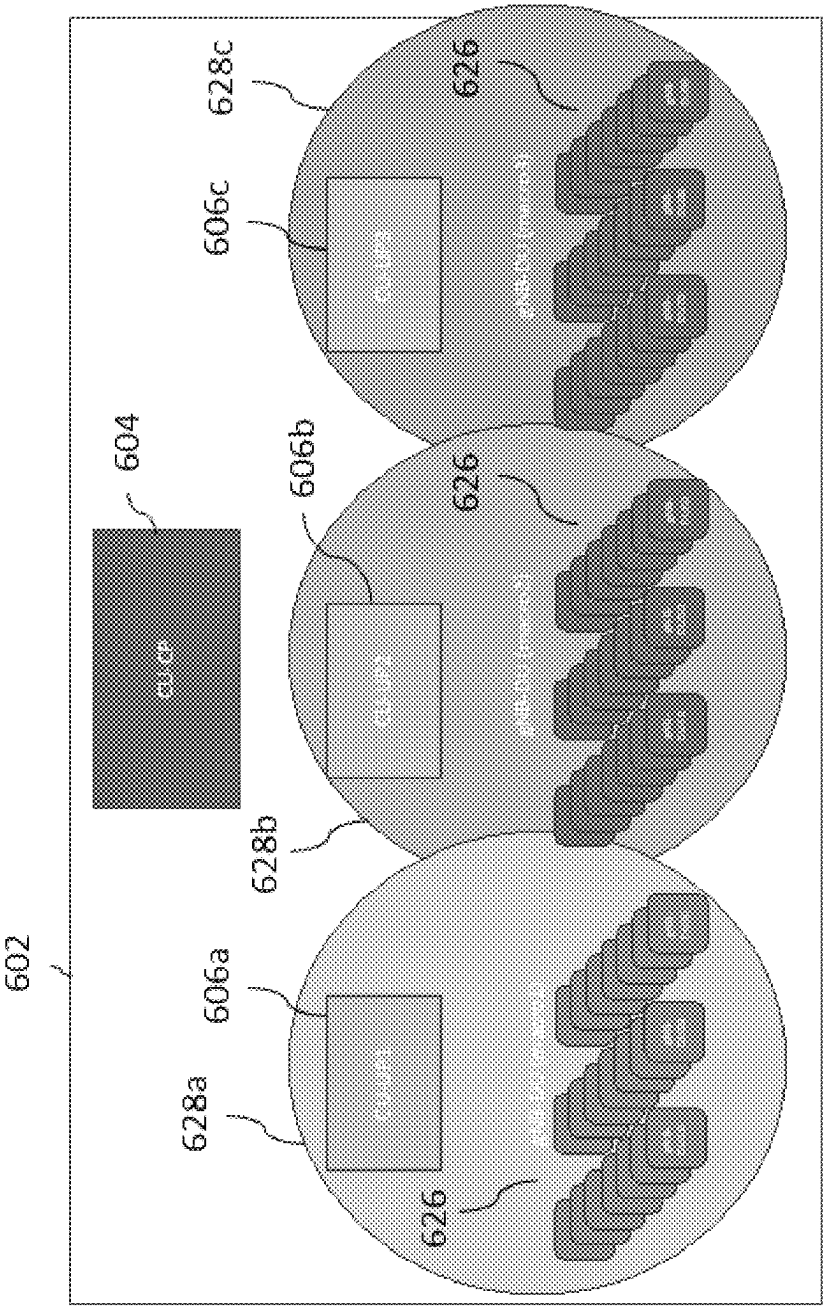


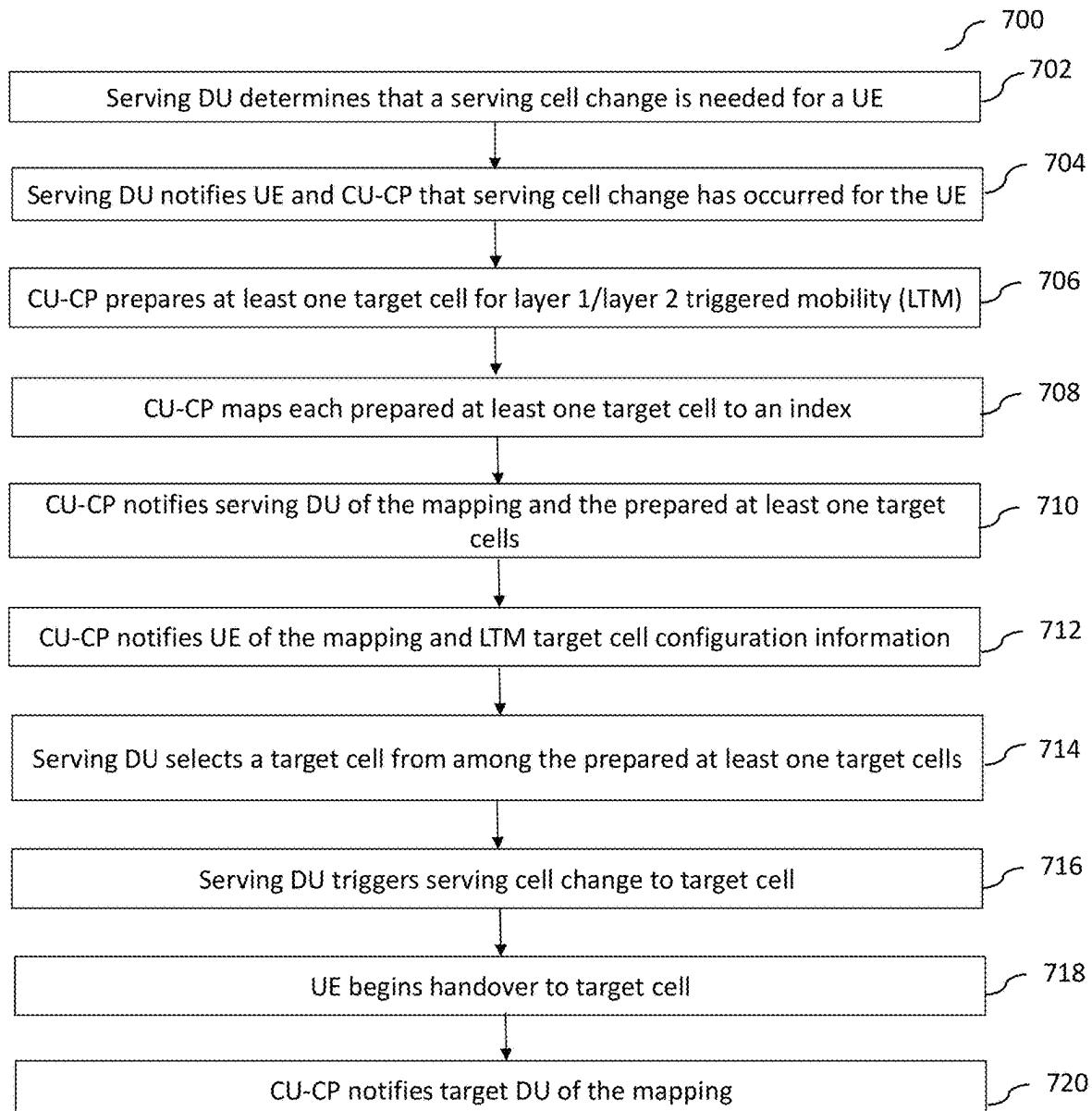
FIG. 7.

FIG. 8.

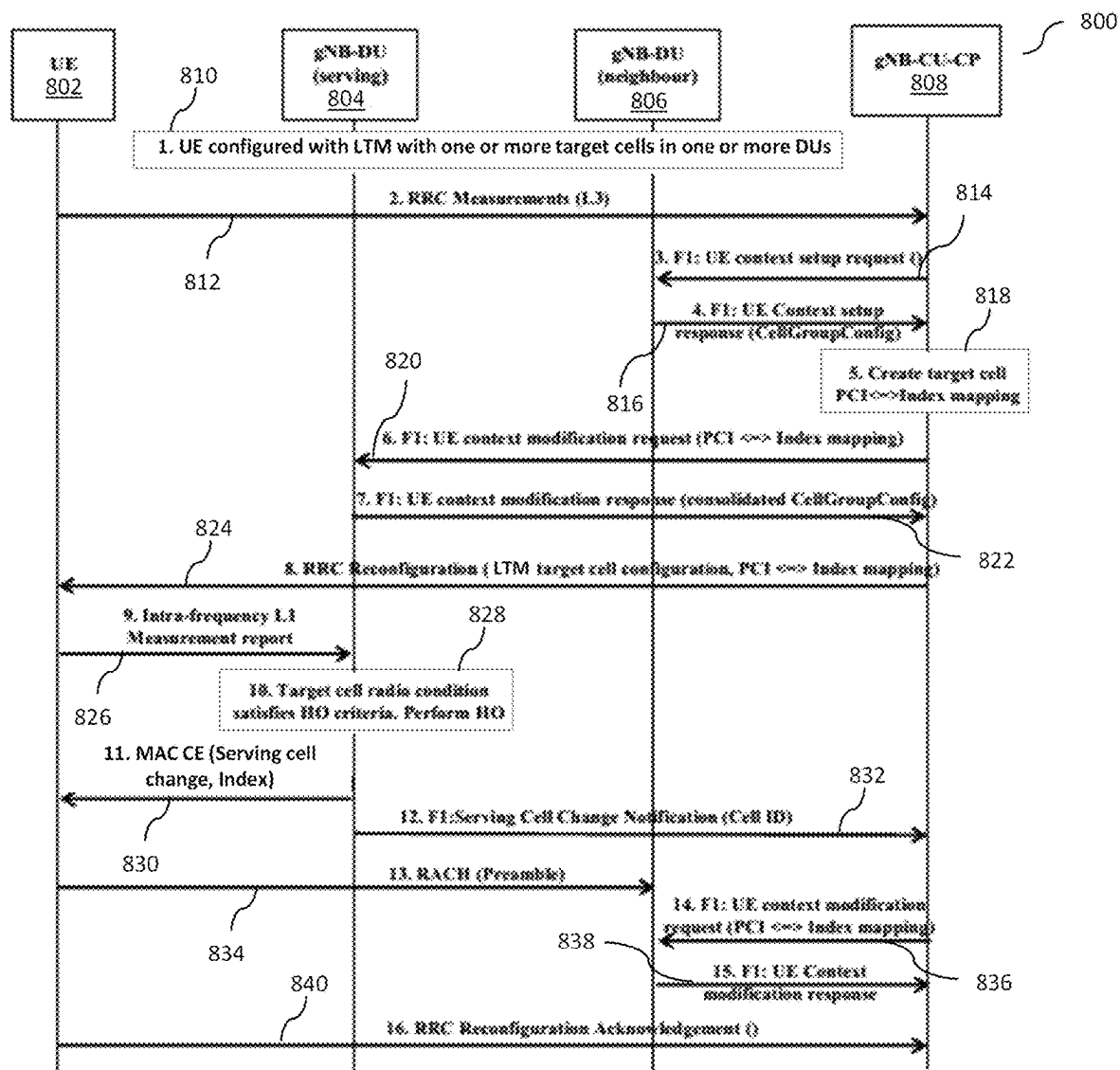


FIG. 9.

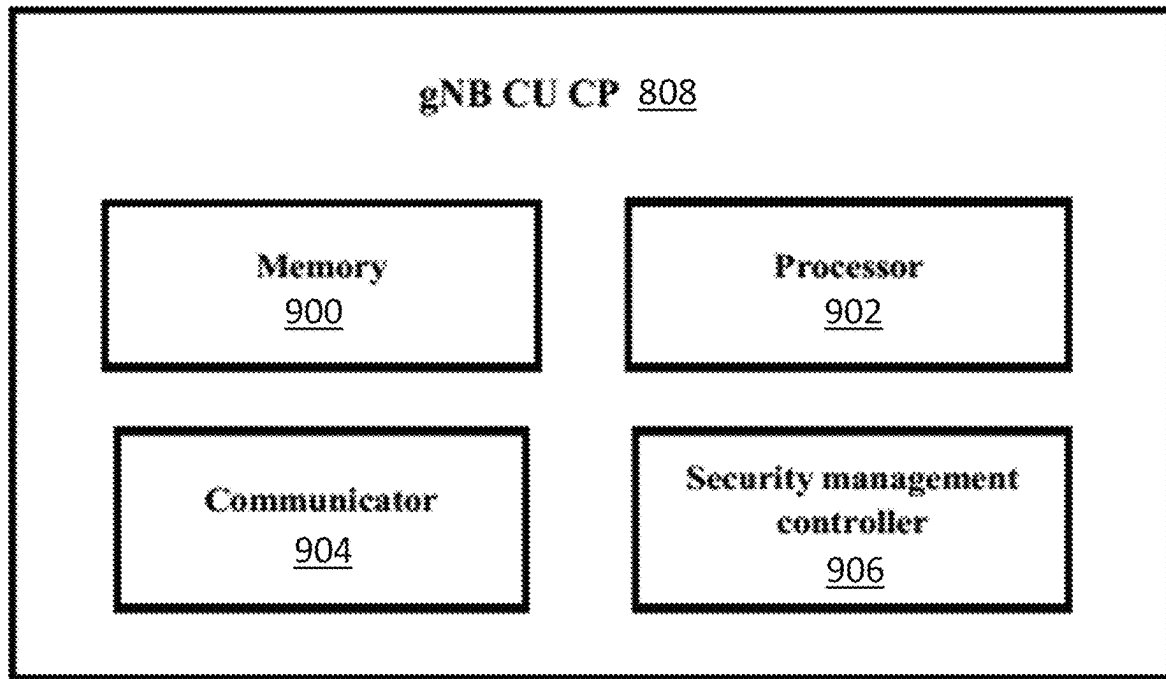


FIG. 10.

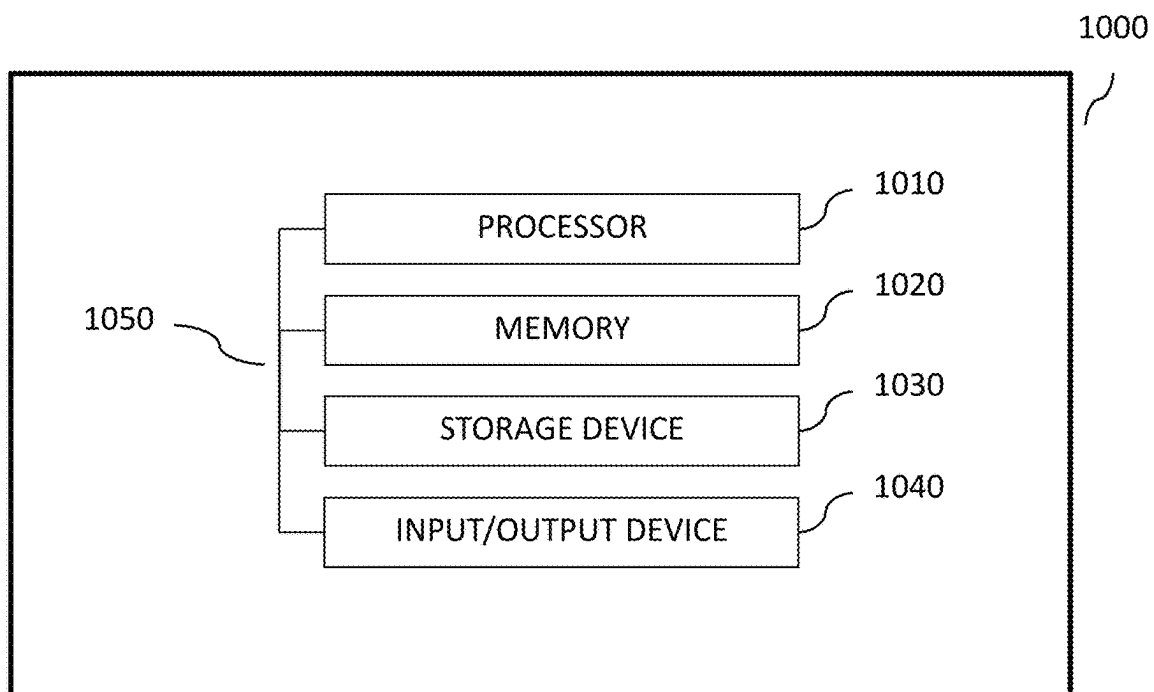
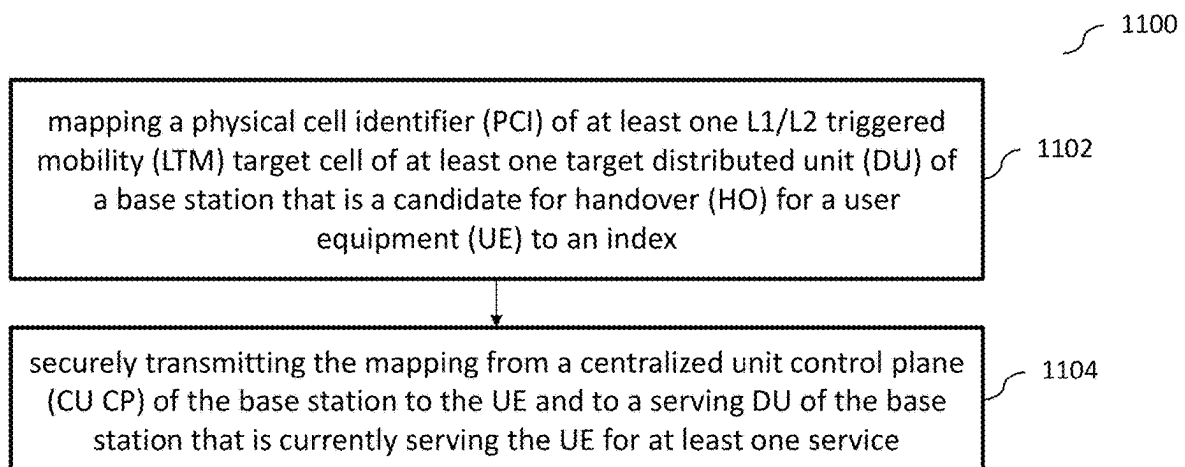


FIG. 11.

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SECURE L1/L2 CENTRIC INTER-CELL MOBILITY EXECUTION

CROSS REFERENCE TO RELATED APPLICATIONS

The present application claims priority to and is the 35 U.S.C. § 371 United States National Phase application based on International Patent Application No. PCT/US23/63090, filed Feb. 23, 2023, and entitled “SECURE L1/L2 CENTRIC INTER-CELL MOBILITY EXECUTION” which claims priority to Indian Patent Application number 202221061428 entitled “Method and system for secure L1/L2 centric inter-cell mobility execution” filed Oct. 28, 2022, which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

In some implementations, the current subject matter relates to telecommunications systems, and in particular, to secure layer 1/layer 2 (L1/L2) centric inter-cell mobility execution.

BACKGROUND

In today’s world, cellular networks provide on-demand communications capabilities to individuals and business entities. Typically, a cellular network is a wireless network that can be distributed over land areas, which are called cells. Each such cell is served by at least one fixed-location transceiver, which is referred to as a cell site or a base station. Each cell can use a different set of frequencies than its neighbor cells in order to avoid interference and provide improved service within each cell. When cells are joined together, they provide radio coverage over a wide geographic area, which enables a large number of mobile telephones, and/or other wireless devices or portable transceivers to communicate with each other and with fixed transceivers and telephones anywhere in the network. Such communications are performed through base stations and are accomplished even if the mobile transceivers are moving through more than one cell during transmission. Major wireless communications providers have deployed such cell sites throughout the world, thereby allowing communications mobile phones and mobile computing devices to be connected to the public switched telephone network and public Internet.

A mobile telephone is a portable telephone that is capable of receiving and/or making telephone and/or data calls through a cell site or a transmitting tower by using radio waves to transfer signals to and from the mobile telephone. In view of a large number of mobile telephone users, current mobile telephone networks provide a limited and shared resource. In that regard, cell sites and handsets can change frequency and use low power transmitters to allow simultaneous usage of the networks by many callers with less interference. Coverage by a cell site can depend on a particular geographical location and/or a number of users that can potentially use the network. For example, in a city, a cell site can have a range of up to approximately ½ mile; in rural areas, the range can be as much as 5 miles; and in some areas, a user can receive signals from a cell site 25 miles away.

The following are examples of some of the digital cellular technologies that are in use by the communications providers: Global System for Mobile Communications (“GSM”),

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General Packet Radio Service (“GPRS”), cdmaOne, CDMA2000, Evolution-Data Optimized (“EV-DO”), Enhanced Data Rates for GSM Evolution (“EDGE”), Universal Mobile Telecommunications System (“UMTS”), Digital Enhanced Cordless Telecommunications (“DECT”), Digital AMPS (“IS-136/TDMA”), and Integrated Digital Enhanced Network (“iDEN”). The Long Term Evolution, or 4G LTE, which was developed by the Third Generation Partnership Project (“3GPP”) standards body, is a standard for a wireless communication of high-speed data for mobile phones and data terminals. A 5G standard is currently being developed and deployed. 3GPP cellular technologies like LTE and 5G NR are evolutions of earlier generation 3GPP technologies like the GSM/EDGE and UMTS/HSPA digital cellular technologies and allows for increasing capacity and speed by using a different radio interface together with core network improvements.

Cellular networks can be divided into radio access networks and core networks. The radio access network (RAN) can include network functions that can handle radio layer communications processing. The core network can include network functions that can handle higher layer communications, e.g., internet protocol (IP), transport layer and applications layer. In some cases, the RAN functions can be split into baseband unit functions and the radio unit functions, where a radio unit connected to a baseband unit via a fronthaul network, for example, can be responsible for lower layer processing of a radio physical layer while a baseband unit can be responsible for the higher layer radio protocols, e.g., MAC, RLC, etc.

A base station for a 5G cellular network can include a centralized unit (CU), one or more distributed units (DUs) communicatively coupled to the CU, and one or more radio units (RUs) each communicatively coupled to at least one of the one or more DUs and each configured to be communicatively coupled to one or more mobile phones and/or other user equipments (UEs). The CU can be logically split into a control plane portion (CU-CP) and one or more user plane portions (CU-UP). During the course of a UE’s communicative coupling with the base station, the DU supporting the UE may change. To achieve such a handover from one DU to another DU, an unsecure message may be sent to the UE **802**, thereby giving rise to a chance of a security breach that can undesirably allow a user’s path to be tracked, or a user’s location to be traced.

SUMMARY

In some implementations, the current subject matter relates to a computer-implemented method. The method can include mapping a physical cell identifier (PCI) of at least one L1/L2 triggered mobility (LTM) target cell of at least one target distributed unit (DU) of a base station that is a candidate for handover (HO) for a user equipment (UE) to an index, and securely transmitting the mapping from a centralized unit control plane (CU-CP) of the base station to the UE and to a serving DU of the base station that is currently serving the UE for at least one service.

The method may allow execution of L1/L2 centric inter-cell mobility without any security compromise.

In some implementations, the current subject matter can include one or more of the following optional features.

In some implementations, the index for each of the at least one LTM targets cells can be unique in the mapping such that each index is uniquely associated with one of the at least one LTM target cells by PCI.

In some implementations, the method can also include securely transmitting the mapping from the CU-CP to a target DU including a one of the at least one LTM targets cells selected for the HO.

In some implementations, the mapping can be transmitted to the UE in a radio resource control (RRC) reconfiguration message, and the mapping can be transmitted to the serving DU in an F1 UE context modification request message.

In some implementations, the method can also include updating the mapping in response to at least one of: one of the at least one LTM targets cells being removed for the UE, and one of the at least one LTM targets cells being replaced for the UE, and the method can also include securely transmitting the updated mapping from the CU-CP to the UE and the serving DU.

In some implementations, the method can also include mapping a PCI of at least one LTM target cell of the at least one target DU that is a candidate for HO for a second UE to an index such that a second mapping is created, and securely transmitting the second mapping from the CU-CP to the second UE.

In some implementations, the serving DU can utilize the index received from the CU-CP when transmitting an LTM HO command to the UE.

In some implementations, each index can be one of numerical, alphabetical, and alphanumeric.

In some implementations, the method can also include storing the mapping in the at least one non-transitory storage media.

In some implementations, the base station can include a Next Generation Radio Access network (NG-RAN) node. Further, the NG-RAN node can include a gNodeB or an ng-eNodeB.

In some implementations, the base station can include the at least one processor and the at least one non-transitory storage media. Further, the CU-CP can include the at least one processor and the at least one non-transitory storage media.

Non-transitory computer program products (i.e., physically embodied computer program products) are also described that store instructions, which when executed by one or more data processors of one or more computing systems, causes at least one data processor to perform operations herein. Similarly, computer systems are also described that may include one or more data processors and memory coupled to the one or more data processors. The memory may temporarily or permanently store instructions that cause at least one processor to perform one or more of the operations described herein. In addition, methods can be implemented by one or more data processors either within a single computing system or distributed among two or more computing systems. Such computing systems can be connected and can exchange data and/or commands or other instructions or the like via one or more connections, including but not limited to a connection over a network (e.g., the Internet, a wireless wide area network, a local area network, a wide area network, a wired network, or the like), via a direct connection between one or more of the multiple computing systems, etc.

The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and advantages of the subject matter described herein will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of this specification, show certain

aspects of the subject matter disclosed herein and, together with the description, help explain some of the principles associated with the disclosed implementations. In the drawings,

FIG. 1a illustrates an exemplary conventional long term evolution ("LTE") communications system:

FIG. 1b illustrates further detail of the exemplary LTE system shown in FIG. 1a:

FIG. 1c illustrates additional detail of the evolved packet core of the exemplary LTE system shown in FIG. 1a:

FIG. 1d illustrates an exemplary evolved Node B of the exemplary LTE system shown in FIG. 1a:

FIG. 2 illustrates further detail of an evolved Node B shown in FIGS. 1a-d:

FIG. 3 illustrates an exemplary virtual radio access network, according to some implementations of the current subject matter:

FIG. 4 illustrates an exemplary 3GPP split architecture to provide its users with use of higher frequency bands:

FIG. 5a illustrates an exemplary 5G wireless communications system:

FIG. 5b illustrates an exemplary layer architecture of the split gNB and/or a split ng-eNB (e.g., next generation eNB) that may be connected to 5GC):

FIG. 5c illustrates an exemplary functional split in the gNB architecture shown in FIGS. 5a-b:

FIG. 6a illustrates an exemplary system, according to some implementations of the current subject matter:

FIG. 6b illustrates an exemplary alternate configuration of the system of FIG. 6a, according to some implementations of the current subject matter:

FIG. 7 illustrates an exemplary method, according to some implementations of the current subject matter:

FIG. 8 illustrates an exemplary signalling diagram, according to some implementations of the current subject matter:

FIG. 9 illustrates an exemplary architecture of a gNB-CU-CP in the signalling diagram of FIG. 8, according to some implementations of the current subject matter:

FIG. 10 illustrates an exemplary system, according to some implementations of the current subject matter: and

FIG. 11 illustrates an exemplary method, according to some implementations of the current subject matter.

DETAILED DESCRIPTION

The current subject matter can provide for systems and methods that can be implemented in wireless communications systems. Such systems can include various wireless communications systems, including 5G New Radio communications systems, long term evolution communication systems, etc.

In general, the current subject matter relates to secure layer 1/layer 2 (L1/L2) centric inter-cell mobility execution.

In some implementations of the current subject matter, a gNB-CU-CP (a control plane portion of a centralized unit of a gNB (gNodeB)) is configured to create indexes for all cells configured as candidate L1/L2 centric inter-cell mobility targets. Currently, in accordance with 3GPP standards (RAN2 agreement), a maximum of eight layer 1/layer 2 triggered mobility (LTM) target cells can be prepared for a given user equipment (UE). The gNB-CU-CP is configured to store mapping between target cell physical cell identifier (PCI) and index of the target cell for future use. Execution of the L1/L2 centric inter-cell mobility may therefore be executed without any security compromise.

3GPP standards may be related to one or more aspects of the current subject matter. Standards of the O-RAN Alliance may also be related to one or more aspects of the current subject matter.

One or more aspects of the current subject matter can be incorporated into transmitter and/or receiver components of base stations (e.g., gNodeBs, eNodeBs, etc.) in such communications systems. The following is a general discussion of long-term evolution communications systems and 5G New Radio communication systems.

I. Long Term Evolution Communications System

FIGS. 1a-c and 2 illustrate an exemplary conventional long-term evolution (“LTE”) communication system 100 along with its various components. An LTE system or a 4G LTE, as it is commercially known, is governed by a standard for wireless communication of high-speed data for mobile telephones and data terminals. The standard is an evolution of the GSM/EDGE (“Global System for Mobile Communications”/“Enhanced Data rates for GSM Evolution”) as well as UMTS/HSPA (“Universal Mobile Telecommunications System”/“High Speed Packet Access”) network technologies. The standard was developed by the 3GPP (“3rd Generation Partnership Project”).

As shown in FIG. 1a, the system 100 can include an evolved universal terrestrial radio access network (“EUTRAN”) 102, an evolved packet core (“EPC”) 108, and a packet data network (“PDN”) 101, where the EUTRAN 102 and EPC 108 provide communication between a user equipment 104 and the PDN 101. The EUTRAN 102 can include a plurality of evolved node B’s (“eNodeB” or “ENODEB” or “enodeb” or “eNB”) or base stations 106 (a, b, c) (as shown in FIG. 1b) that provide communication capabilities to a plurality of user equipment 104(a, b, c). The user equipment 104 can be a mobile telephone, a smartphone, a tablet, a personal computer, a personal digital assistant (“PDA”), a server, a data terminal, and/or any other type of user equipment, and/or any combination thereof. The user equipment 104 can connect to the EPC 108 and eventually, the PDN 101, via any eNodeB 106. Typically, the user equipment 104 can connect to the nearest, in terms of distance, eNodeB 106. In the LTE system 100, the EUTRAN 102 and EPC 108 work together to provide connectivity, mobility and services for the user equipment 104.

FIG. 1b illustrates further detail of the network 100 shown in FIG. 1a. As stated above, the EUTRAN 102 includes a plurality of eNodeBs 106, also known as cell sites. The eNodeBs 106 provides radio functions and performs key control functions including scheduling of air link resources or radio resource management, active mode mobility or handover, and admission control for services. The eNodeBs 106 are responsible for selecting which mobility management entities (MMEs, as shown in FIG. 1c) will serve the user equipment 104 and for protocol features like header compression and encryption. The eNodeBs 106 that make up an EUTRAN 102 collaborate with one another for radio resource management and handover.

Communication between the user equipment 104 and the eNodeB 106 occurs via an air interface 122 (also known as “LTE-Uu” interface). As shown in FIG. 1b, the air interface 122 provides communication between user equipment 104b and the eNodeB 106a. The air interface 122 uses Orthogonal Frequency Division Multiple Access (“OFDMA”) and Single Carrier Frequency Division Multiple Access (“SC-FDMA”), an OFDMA variant, on the downlink and uplink respectively. OFDMA allows use of multiple known antenna techniques, such as, Multiple Input Multiple Output (“MIMO”).

The air interface 122 uses various protocols, which include a radio resource control (“RRC”) for signaling between the user equipment 104 and eNodeB 106 and non-access stratum (“NAS”) for signaling between the user equipment 104 and MME (as shown in FIG. 1c). In addition to signaling, user traffic is transferred between the user equipment 104 and eNodeB 106. Both signaling and traffic in the system 100 are carried by physical layer (“PHY”) channels.

Multiple eNodeBs 106 can be interconnected with one another using an X2 interface 130(a, b, c). As shown in FIG. 1b, X2 interface 130a provides interconnection between eNodeB 106a and eNodeB 106b; X2 interface 130b provides interconnection between eNodeB 106a and eNodeB 106c; and X2 interface 130c provides interconnection between eNodeB 106b and eNodeB 106c. The X2 interface can be established between two eNodeBs in order to provide an exchange of signals, which can include a load- or interference-related information as well as handover-related information. The eNodeBs 106 communicate with the evolved packet core 108 via an S1 interface 124(a, b, c). The S1 interface 124 can be split into two interfaces: one for the control plane (shown as control plane interface (S1-MME interface) 128 in FIG. 1c) and the other for the user plane (shown as user plane interface (S1-U interface) 125 in FIG. 1c).

The EPC 108 establishes and enforces Quality of Service (“QoS”) for user services and allows user equipment 104 to maintain a consistent internet protocol (“IP”) address while moving. It should be noted that each node in the network 100 has its own IP address. The EPC 108 is designed to inter-work with legacy wireless networks. The EPC 108 is also designed to separate control plane (i.e., signaling) and user plane (i.e., traffic) in the core network architecture, which allows more flexibility in implementation, and independent scalability of the control and user data functions.

The EPC 108 architecture is dedicated to packet data and is shown in more detail in FIG. 1c. The EPC 108 includes a serving gateway (S-GW) 110, a PDN gateway (P-GW) 112, a mobility management entity (“MME”) 114, a home subscriber server (“HSS”) 116 (a subscriber database for the EPC 108), and a policy control and charging rules function (“PCRF”) 118. Some of these (such as S-GW, P-GW, MME, and HSS) are often combined into nodes according to the manufacturer’s implementation.

The S-GW 110 functions as an IP packet data router and is the user equipment’s bearer path anchor in the EPC 108. Thus, as the user equipment moves from one eNodeB 106 to another during mobility operations, the S-GW 110 remains the same and the bearer path towards the EUTRAN 102 is switched to talk to the new eNodeB 106 serving the user equipment 104. If the user equipment 104 moves to the domain of another S-GW 110, the MME 114 will transfer all of the user equipment’s bearer paths to the new S-GW. The S-GW 110 establishes bearer paths for the user equipment to one or more P-GWs 112. If downstream data are received for an idle user equipment, the S-GW 110 buffers the downstream packets and requests the MME 114 to locate and reestablish the bearer paths to and through the EUTRAN 102.

The P-GW 112 is the gateway between the EPC 108 (and the user equipment 104 and the EUTRAN 102) and PDN 101 (shown in FIG. 1a). The P-GW 112 functions as a router for user traffic as well as performs functions on behalf of the user equipment. These include IP address allocation for the user equipment, packet filtering of downstream user traffic to ensure it is placed on the appropriate bearer path, enforce-

ment of downstream Qos, including data rate. Depending upon the services a subscriber is using, there may be multiple user data bearer paths between the user equipment **104** and P-GW **112**. The subscriber can use services on PDNs served by different P-GWs, in which case the user equipment has at least one bearer path established to each P-GW **112**. During handover of the user equipment from one eNodeB to another, if the S-GW **110** is also changing, the bearer path from the P-GW **112** is switched to the new S-GW.

The MME **114** manages user equipment **104** within the EPC **108**, including managing subscriber authentication, maintaining a context for authenticated user equipment **104**, establishing data bearer paths in the network for user traffic, and keeping track of the location of idle mobiles that have not detached from the network. For idle user equipment **104** that needs to be reconnected to the access network to receive downstream data, the MME **114** initiates paging to locate the user equipment and re-establishes the bearer paths to and through the EUTRAN **102**. MME **114** for a particular user equipment **104** is selected by the eNodeB **106** from which the user equipment **104** initiates system access. The MME is typically part of a collection of MMEs in the EPC **108** for the purposes of load sharing and redundancy. In the establishment of the user's data bearer paths, the MME **114** is responsible for selecting the P-GW **112** and the S-GW **110**, which will make up the ends of the data path through the EPC **108**.

The PCRF **118** is responsible for policy control decision-making, as well as for controlling the flow-based charging functionalities in the policy control enforcement function ("PCEF"), which resides in the P-GW **110**. The PCRF **118** provides the Qos authorization (QOS class identifier ("QCI") and bit rates) that decides how a certain data flow will be treated in the PCEF and ensures that this is in accordance with the user's subscription profile.

As stated above, the IP services **119** are provided by the PDN **101** (as shown in FIG. **1a**).

FIG. **1d** illustrates an exemplary structure of eNodeB **106**. The eNodeB **106** can include at least one remote radio head ("RRH") **132** (typically, there can be three RRH **132**) and a baseband unit ("BBU") **134**. The RRH **132** can be connected to antennas **136**. The RRH **132** and the BBU **134** can be connected using an optical interface that is compliant with common public radio interface ("CPRI")/enhanced CPRI ("eCPRI") **142** standard specification either using RRH specific custom control and user plane framing methods or using O-RAN Alliance compliant Control and User plane framing methods. The operation of the eNodeB **106** can be characterized using the following standard parameters (and specifications): radio frequency band (Band4, Band9, Band17, etc.) bandwidth (5, 10, 15, 20 MHz), access scheme (downlink: OFDMA; uplink: SC-OFDMA), antenna technology (Single user and multi user MIMO; Uplink: Single user and multi user MIMO), number of sectors (6 maximum), maximum transmission rate (downlink: 150 Mb/s; uplink: 50 Mb/s), S1/X2 interface (1000Base-SX, 1000Base-T), and mobile environment (up to 350 km/h). The BBU **134** can be responsible for digital baseband signal processing, termination of S1 line, termination of X2 line, call processing and monitoring control processing. IP packets that are received from the EPC **108** (not shown in FIG. **1d**) can be modulated into digital baseband signals and transmitted to the RRH **132**. Conversely, the digital baseband signals received from the RRH **132** can be demodulated into IP packets for transmission to EPC **108**.

The RRH **132** can transmit and receive wireless signals using antennas **136**. The RRH **132** can convert (using converter ("CONV") **140**) digital baseband signals from the BBU **134** into radio frequency ("RF") signals and power amplify (using amplifier ("AMP") **138**) them for transmission to user equipment **104** (not shown in FIG. **1d**). Conversely, the RF signals that are received from user equipment **104** are amplified (using AMP **138**) and converted (using CONV **140**) to digital baseband signals for transmission to the BBU **134**.

FIG. **2** illustrates an additional detail of an exemplary eNodeB **106**. The eNodeB **106** includes a plurality of layers: LTE layer 1 **202**, LTE layer 2 **204**, and LTE layer 3 **206**. The LTE layer 1 includes a physical layer ("PHY"). The LTE layer 2 includes a medium access control ("MAC"), a radio link control ("RLC"), a packet data convergence protocol ("PDCP"). The LTE layer 3 includes various functions and protocols, including a radio resource control ("RRC"), a dynamic resource allocation, eNodeB measurement configuration and provision, a radio admission control, a connection mobility control, and radio resource management ("RRM"). The RLC protocol is an automatic repeat request ("ARQ") fragmentation protocol used over a cellular air interface. The RRC protocol handles control plane signaling of LTE layer 3 between the user equipment and the EUTRAN. RRC includes functions for connection establishment and release, broadcast of system information, radio bearer establishment/reconfiguration and release. RRC connection mobility procedures, paging notification and release, and outer loop power control. The PDCP performs IP header compression and decompression, transfer of user data and maintenance of sequence numbers for Radio Bearers. The BBU **134**, shown in FIG. **1d**, can include LTE layers L1-L3.

One of the primary functions of the eNodeB **106** is radio resource management, which includes scheduling of both uplink and downlink air interface resources for user equipment **104**, control of bearer resources, and admission control. The eNodeB **106**, as an agent for the EPC **108**, is responsible for the transfer of paging messages that are used to locate mobiles when they are idle. The eNodeB **106** also communicates common control channel information over the air, header compression, encryption and decryption of the user data sent over the air, and establishing handover reporting and triggering criteria. As stated above, the eNodeB **106** can collaborate with other eNodeB **106** over the X2 interface for the purposes of handover and interference management. The eNodeBs **106** communicate with the EPC's MME via the S1-MME interface and to the S-GW with the S1-U interface. Further, the eNodeB **106** exchanges user data with the S-GW over the S1-U interface. The eNodeB **106** and the EPC **108** have a many-to-many relationship to support load sharing and redundancy among MMEs and S-GWs. The eNodeB **106** selects an MME from a group of MMEs so the load can be shared by multiple MMEs to avoid congestion.

II. 5G NR Wireless Communications Networks

In some implementations, the current subject matter relates to a 5G new radio ("NR") communications system. The 5G NR is a next telecommunications standard beyond the 4G/IMT-Advanced standards. 5G networks offer at higher capacity than current 4G, allow higher number of mobile broadband users per area unit, and allow consumption of higher and/or unlimited data quantities in gigabyte per month and user. This can allow users to stream high-definition media many hours per day using mobile devices, even when it is not possible to do so with Wi-Fi networks. 5G networks have an improved support of device-to-device

communication, lower cost, lower latency than 4G equipment and lower battery consumption, etc. Such networks have data rates of tens of megabits per second for a large number of users, data rates of 100 Mb/s for metropolitan areas. 1 Gb/s simultaneously to users within a confined area (e.g., office floor), a large number of simultaneous connections for wireless sensor networks, an enhanced spectral efficiency, improved coverage, enhanced signaling efficiency. 1-10 ms latency, reduced latency compared to existing systems.

FIG. 3 illustrates an exemplary virtual radio access network 300. The network 300 can provide communications between various components, including a base station (e.g., eNodeB, gNodeB) 301, a radio equipment 303, a centralized unit 302, a digital unit 304, and a radio device 306. The components in the system 300 can be communicatively coupled to a core using a backhaul link 305. A centralized unit ("CU") 302 can be communicatively coupled to a distributed unit ("DU") 304 using a midhaul connection 308. The radio frequency ("RU") components 306 can be communicatively coupled to the DU 304 using a fronthaul connection 310.

In some implementations, the CU 302 can provide intelligent communication capabilities to one or more DU units 304. The units 302, 304 can include one or more base stations, macro base stations, micro base stations, remote radio heads, etc. and/or any combination thereof.

In lower layer split architecture environment, a CPRI bandwidth requirement for NR can be 100s of Gb/s. CPRI compression can be implemented in the DU and RU (as shown in FIG. 3). In 5G communications systems, compressed CPRI over Ethernet frame is referred to as eCPRI and is the recommended fronthaul network. The architecture can allow for standardization of fronthaul/midhaul, which can include a higher layer split (e.g., Option 2 or Option 3-1 (Upper/Lower RLC split architecture)) and fronthaul with L1-split architecture (Option 7).

In some implementations, the lower layer-split architecture (e.g., Option 7) can include a receiver in the uplink, joint processing across multiple transmission points (TPs) for both DL/UL, and transport bandwidth and latency requirements for ease of deployment. Further, the current subject matter's lower layer-split architecture can include a split between cell-level and user-level processing, which can include cell-level processing in remote unit ("RU") and user-level processing in DU. Further, using the current subject matter's lower layer-split architecture, frequency-domain samples can be transported via Ethernet fronthaul, where the frequency-domain samples can be compressed for reduced fronthaul bandwidth.

FIG. 4 illustrates an exemplary communications system 400 that can implement a 5G technology and can provide its users with use of higher frequency bands (e.g., greater than 10 GHz). The system 400 can include a macro cell 402 and small cells 404, 406.

A mobile device 408 can be configured to communicate with one or more of the small cells 404, 406. The system 400 can allow splitting of control planes (C-plane) and user planes (U-plane) between the macro cell 402 and small cells 404, 406, where the C-plane and U-plane are utilizing different frequency bands. In particular, the small cells 404, 406 can be configured to utilize higher frequency bands when communicating with the mobile device 408. The macro cell 402 can utilize existing cellular bands for C-plane communications. The mobile device 408 can be communicatively coupled via U-plane 412, where the small cell (e.g., small cell 406) can provide higher data rate and more

flexible/cost/energy efficient operations. The macro cell 402, via C-plane 410, can maintain good connectivity and mobility. Further, in some cases, LTE and NR can be transmitted on the same frequency.

FIG. 5a illustrates an exemplary 5G wireless communication system 500, according to some implementations of the current subject matter. The system 500 can be configured to have a lower layer split architecture in accordance with Option 7-2. The system 500 can include a core network 502 (e.g., 5G Core) and one or more gNodeBs (or gNBs), where the gNBs can have a centralized unit gNB-CU. The gNB-CU can be logically split into control plane portion, gNB-CU-CP, 504 and one or more user plane portions, gNB-CU-UP, 506. The control plane portion 504 and the user plane portion 506 can be configured to be communicatively coupled using an E1 communication interface 514 (as specified in the 3GPP Standard). The control plane portion 504 can be configured to be responsible for execution of the RRC and PDCP protocols of the radio stack.

The control plane and user plane portions 504, 506 of the centralized unit of the gNB can be configured to be communicatively coupled to one or more distributed units (DU) 508, 510, in accordance with the higher layer split architecture. The distributed units 508, 510 can be configured to execute RLC, MAC and upper part of PHY layers protocols of the radio stack. The control plane portion 504 can be configured to be communicatively coupled to the distributed units 508, 510 using F1-C communication interfaces 516, and the user plane portions 506 can be configured to be communicatively coupled to the distributed units 508, 510 using F1-U communication interfaces 518. The distributed units 508, 510 can be coupled to one or more remote radio units (RU) 512 via a fronthaul network 520 (which may include one or switches, links, etc.), which in turn communicate with one or more user equipment (not shown in FIG. 5a). The remote radio units 512 can be configured to execute a lower part of the PHY layer protocols as well as provide antenna capabilities to the remote units for communication with user equipments (similar to the discussion above in connection with FIGS. 1a-2).

FIG. 5b illustrates an exemplary layer architecture 530 of the split gNB. The architecture 530 can be implemented in the communications system 500 shown in FIG. 5a, which can be configured as a virtualized disaggregated radio access network (RAN) architecture, whereby layers L1, L2, L3 and radio processing can be virtualized and disaggregated in the centralized unit(s), distributed unit(s) and radio unit(s). As shown in FIG. 5b, the gNB-DU 508 can be communicatively coupled to the gNB-CU-CP control plane portion 504 (also shown in FIG. 5a) and gNB-CU-UP user plane portion 506. Each of components 504, 506, 508 can be configured to include one or more layers.

The gNB-DU 508 can include RLC, MAC, and PHY layers as well as various communications sublayers. These can include an F1 application protocol (F1-AP) sublayer, a GPRS tunneling protocol (GTP-U) sublayer, a stream control transmission protocol (SCTP) sublayer, a user datagram protocol (UDP) sublayer and an internet protocol (IP) sublayer. As stated above, the distributed unit 508 may be communicatively coupled to the control plane portion 504 of the centralized unit, which may also include F1-AP, SCTP, and IP sublayers as well as radio resource control, and PDCP-control (PDCP-C) sublayers. Moreover, the distributed unit 508 may also be communicatively coupled to the user plane portion 506 of the centralized unit of the gNB.

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The user plane portion **506** may include service data adaptation protocol (SDAP). PDCP-user (PDCP-U). GTP-U, UDP, and IP sublayers.

FIG. **5c** illustrates an exemplary functional split in the gNB architecture shown in FIGS. **5a-b**. As shown in FIG. **5c**, the gNB-DU **508** may be communicatively coupled to the gNB-CU-CP **504** and gNB-CU-UP **506** using an F1-C communication interface. The gNB-CU-CP **504** and gNB-CU-UP **506** may be communicatively coupled using an E1 communication interface. The higher part of the PHY layer (or Layer 1) may be executed by the gNB-DU **508**, whereas the lower parts of the PHY layer may be executed by the RUs (not shown in FIG. **5c**). As shown in FIG. **5c**, the RRC and PDCP-C portions may be executed by the control plane portion **504**, and the SDAP and PDCP-U portions may be executed by the user plane portion **506**.

Some of the functions of the PHY layer in 5G communications network can include error detection on the transport channel and indication to higher layers. FEC encoding/decoding of the transport channel, hybrid ARQ soft-combining, rate matching of the coded transport channel to physical channels, mapping of the coded transport channel onto physical channels, power weighting of physical channels, modulation and demodulation of physical channels, frequency and time synchronization, radio characteristics measurements and indication to higher layers, MIMO antenna processing, digital and analog beamforming, RF processing, as well as other functions.

The MAC sublayer of Layer 2 can perform beam management, random access procedure, mapping between logical channels and transport channels, concatenation of multiple MAC service data units (SDUs) belonging to one logical channel into transport block (TB), multiplexing/demultiplexing of SDUs belonging to logical channels into/from TBs delivered to/from the physical layer on transport channels, scheduling information reporting, error correction through HARQ, priority handling between logical channels of one UE, priority handling between UEs by means of dynamic scheduling, transport format selection, and other functions. The RLC sublayer's functions can include transfer of upper layer packet data units (PDUs), error correction through ARQ, reordering of data PDUs, duplicate and protocol error detection, re-establishment, etc. The PDCP sublayer can be responsible for transfer of user data, various functions during re-establishment procedures, retransmission of SDUs, SDU discard in the uplink, transfer of control plane data, and others.

Layer 3's RRC sublayer can perform broadcasting of system information to NAS and AS, establishment, maintenance and release of RRC connection, security, establishment, configuration, maintenance and release of point-point radio bearers, mobility functions, reporting, and other functions.

III. Secure L1/L2 Centric Inter-Cell Mobility Execution

Several companies in RAN2 have expressed concern regarding security of lower layer mobility (LLM) execution. Since a MAC control element (CE) message in downlink issued by a serving gNB-DU is insecure (not encrypted), there is a chance of a security breach. For example, since a particular MAC CE is associated with the execution of a mobility event, a user's path can be tracked, or a user's location can be traced. In a handover (HO) in which one or more services of a UE are being handed over from a serving cell (also referred to herein as a "source cell") to a target cell, the serving gNB-DU may transmit to the UE a MAC CE that includes a target cell physical cell identifier (PCI), or other identifier, that identifies the target cell to the UE. Because

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the MAC CE message to the UE is insecure, information about the UE and/or the target cell is susceptible to security breach.

Layer 1/layer 2 triggered mobility (LTM) is an updated term for LLM. RAN2 has agreed on a definition of LTM. In general, LTM is a mobility procedure that allows a network to switch a UE from a source cell to a target cell without necessarily requiring a reconfiguration with sync. In particular, the network, based on L1 measurements received, can indicate in an L2 signaling (e.g., a message such as a MAC CE) a beam belonging to an LTM candidate cell to which the UE should perform the LTM cell switch procedure. The UE is provided with at least one LTM candidate cell configuration by the network before execution of an LTM cell switch procedure.

In some implementations of the current subject matter, a gNB-CU-CP is configured to create indexes for all cells configured as candidate L1/L2 centric inter-cell mobility targets. Currently, in accordance with 3GPP standards (RAN2 agreement), a maximum of eight LTM target cells can be prepared for a given UE. The gNB-CU-CP is configured to store mapping between target cell PCI and index of the target cell for future use. Execution of L1/L2 centric inter-cell mobility may therefore be executed without any security compromise.

In some implementations of the current subject matter, a base station (e.g., the gNodeB of FIG. **5a**, a next generation RAN (NG-RAN) node such as an eNodeB or a gNodeB, etc.) of a wireless communication system (e.g., a 5G wireless communication system, a 6G or later generation wireless communication system, etc.) can have a disaggregated architecture in which the base station includes one gNB-CU-CP (e.g., gNB-CU-CP **504** of FIGS. **5a-5c**, etc.) and more than one CU-UP (e.g., gNB-CU-UP **506** of FIGS. **5a-5c**, etc.) and gNB-DU (e.g., gNB-DUs **508**, **510** of FIGS. **5a-5c**, etc.). The base station can be configured to execute secure L1/L2 centric inter-cell mobility when a UE is handed off from one cell (serving cell) of the base station to another cell (target cell) of the base station.

FIG. **6a** illustrates an exemplary system **600** configured to execute secure L1/L2 centric inter-cell mobility. A base station **602** in this illustrated implementation is a gNB configured to be in a 5G wireless communication system similar to the 5G wireless communication system **500** of FIG. **5a** discussed above, but other base stations can be similarly configured and used in executing secure L1/L2 centric inter-cell mobility. In the illustrated implementation of FIG. **6a**, the base station **602** includes a plurality of CU-UPs **606a**, **606b**, **606c**. The base station **602** includes three CU-UPs **606a**, **606b**, **606c** in this illustrated implementation but can include another plural number of CU-UPs. The CU of the base station **602** that includes the plurality of CU-UPs **606a**, **606b**, **606c** is configured to be communicatively coupled with a core network (not shown in FIG. **6a**), e.g., the 5GC **502** of FIG. **5a**, etc.

The CU of the base station **602** also includes a CU-CP **604** configured to be communicatively coupled to the CU's user plane portion **606a**, **606b**, **606c** using an E1 communication interface **614**. The E1 interface **614** includes three communication links in this illustrated implementation to reflect that there are three CU-UPs **606a**, **606b**, **606c** with which the CU-CP **604** can be configured to communicate.

The base station **602** also includes a plurality of DUs **608**, **610**. The base station **602** includes two DUs **608**, **610** in this illustrated implementation but can include another plural number of DUs. The CU-CP **604** is configured to be communicatively coupled to the DUs **608**, **610** using F1-C

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communication interfaces **616**. The CU-UPs **606a**, **606b**, **606c** are configured to be communicatively coupled to the DUs **608**, **610** using F1-U communication interfaces **618**. The F1-U interface **618** associated with each of the DUs **608**, **610** includes three communication links in this illustrated implementation to reflect that there are three CU-UPs **606a**, **606b**, **606c** with which each DU **608**, **610** can be configured to communicate.

The base station **602** also includes a plurality of RUs **612**. The base station **602** includes five RUs **612** in this illustrated implementation but can include another plural number of RUs. The RUs **612** are configured to be communicatively coupled to the DUs **608**, **610** via a fronthaul network **620**. Additionally, each of the RUs **612** is configured to be communicatively coupled to one or more UEs **622**. In this illustrated implementation, two of the RUs **612** are shown communicatively coupled to one UE **622**, two of the RUs **612** are shown communicatively coupled to two UEs **622**, and one of the RUs **612** is shown communicatively coupled to three UEs **622**, but each of the RUs **612** can be coupled to another number of UEs same or different from any of the other RUs **612**.

Secure L1/L2 centric inter-cell mobility execution can be configured to occur when one of the UEs communicatively coupled with the base station **602** is handed off from one of the DUs **608**, **610** of the base station **602** to the other of the DUs **608**, **610** also of the same base station **602**. The one of the DUs **608**, **610** currently providing service to the UE **622** is referred to as a “serving DU” due to it currently providing service to the UE **622**, e.g., currently serving the UE **622**. The one of the DUs **608**, **610** to which the UE’s service is being handed off is referred to as a “target DU” due to it being targeted to provide service to the UE **622**.

A system in which executing secure L1/L2 centric inter-cell mobility can be configured to occur is further described with respect to FIG. **6b**. FIG. **6b** illustrates the CU-CP **604** and the CU-UPs **606a**, **606b**, **606c** of FIG. **6a**, but in the illustrated implementation of FIG. **6b**, the base station **602** includes more than two DUs. In the illustrated implementation of FIG. **6b**, the base station **602** includes sixty-six DUs. Three of the DUs **628a**, **628b**, **628c** are macro cells (labeled macro1, macro2, and macro3 in FIG. **6b**), and sixty-three of the DUs **626** small cells (nine of which are labeled in FIG. **6b** as gNB-DU10, gNB-DU20, gNB-DU30, gNB-DU40, gNB-DU50, gNB-DU60, gNB-DU70, gNB-DU80, and gNB-DU90). The base station **602** can include another number of macro cells and/or another number of small cells. The macro1 DU **628a**, the macro2 DU **628b**, and twenty-one of the small cell DUs **626** including gNB-DU10, gNB-DU20, and gNB-DU30) are configured to be served by the first CU-UP **606a** (labeled CU-UP1 in FIG. **6b**). The macro1 DU **628a**, macro2 DU **628b**, macro3 DU **628c**, and twenty-one of the small cell DUs **626** including gNB-DU40, gNB-DU50, and gNB-DU60) are configured to be served by the second CU-UP **606b** (labeled CU-UP2 in FIG. **6b**). The macro2 DU **628b**, macro3 DU **628c**, and twenty-one of the small cell DUs **626** including gNB-DU70, gNB-DU80, and gNB-DU90) are configured to be served by the third CU-UP **606c** (labeled CU-UP3 in FIG. **6b**).

In the implementation shown in FIG. **6b**, each CU-UP **606a**, **606b**, **606c** is serving a subset of DUs **626**, **628a**, **628b**, **628c** for all the services. However, a CU-UP can serve all the DUs of the base station for one service (e.g., enhanced mobile broadband (eMBB)), while serving a subset of the DUs for another service (e.g., vehicle-to-everything (V2X) or ultra-reliable low latency communication (URLLC)).

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FIG. **7** illustrates an exemplary method **700**, according to some implementations of the current subject matter. The method **700** of FIG. **7** is described with respect to an exemplary system **800** illustrated in FIG. **8** but can be implemented similarly with other systems. e.g., the system **100** of FIGS. **1a-1c** and **2**, the system **400** of FIG. **4**, the system **500** of FIG. **5a**, the systems of FIGS. **6a** and **6b**, etc. The system **800** of FIG. **8** is a 5G system, but as mentioned above, executing secure L1/L2 centric inter-cell mobility as described herein can be performed with other types of wireless communications systems, such as LTE wireless communications systems or 6G or later generation wireless communications systems. Various elements in FIG. **8** are numbered as sequential steps, but such numbering is not intended to indicate that only these numbered steps in this sequential order may be performed in the system **800**. One or more additional steps may be present before and/or after any one or more of the sequentially numbered steps shown in FIG. **8**.

In the system **800**, a UE **802** (e.g., UE **104** of FIGS. **1a-1c**, UE **622** of FIG. **6a**, etc.) is configured **810** with LTM with one or more target cells in one or more DUs **804**, **806** (e.g., DU **508** of FIGS. **5a-5c**, DU **510** of FIG. **5a**, DU **608** of FIG. **6a**, DU **610** of FIG. **6a**, DUs **626** of FIG. **6b**, DUs **628a**, **628b**, **628c** of FIG. **6b**, etc.) of a base station. e.g., a gNB (e.g., gNodeB of FIG. **5a**, gNodeB **602** of FIGS. **6a** and **6b**, etc.). For ease of explanation the system **800** is shown in FIG. **8** with one UE **802** communicatively coupled to the base station and with the base station including two DUs **804**, **806**, but more than one UE can be communicatively coupled to the base station and/or the base station can include more than two DUs. The base station of the system **800** also includes a CU including a CU-CP **808** (e.g., gNB-CU-CP **504** of FIGS. **5a-5c**, CU-CP **604** of FIGS. **6a** and **6b**, etc.) and one or more CU-UPs (e.g., gNB-CU-UP **506** of FIGS. **5a-5c**, CU-UPs **606a**, **606b**, **606c** of FIGS. **6a** and **6b**, etc.) (not shown in FIG. **8**), and a plurality of RUs (e.g., RUs **512** of FIG. **5a**, RUs **612** of FIG. **6a**, etc.) (not shown in FIG. **8**). The UE **802** is currently being served by the serving DU **804**. Additionally, the base station of FIG. **8** is communicatively coupled with a core network (e.g., EPC **108** of FIGS. **1a-1c** and **2**, 5GC **502** of FIG. **5a**, etc.) (not shown in FIG. **8**).

In response to determining **702** that a serving cell change should occur, the serving DU **804** notifies **704** the UE **802** of a serving cell change. The notification **704** to the UE **802** can include the serving DU **804** transmitting a serving cell change command. e.g., a MAC CE, to the UE **802**.

Also in response to determining **702** that a cell service change should occur, the serving DU **804** notifies **704** the CU-CP **808** that that a serving cell change for the UE **802** has occurred. The notification **704** can thus identify the UE **802** to the CU-CP **808**, such as with an identifier, in accordance with 3GPP standards, known to the serving DU **804** that uniquely identifies the UE **802** to the CU-CP **808**. The notification **704** to the CU-CP **808** can include the serving DU **804** transmitting a SERVING CELL CHANGE NOTIFICATION message to the CU-CP **808** using an F1 communication interface. The SERVING CELL CHANGE NOTIFICATION message can include a cell identification (ID) that uniquely identifies the UE **802** that underwent the serving cell change.

In response to receiving the serving cell change command from the serving DU **804**, the UE **802** transmits a Radio Resource Control (RRC) Reconfigure Acknowledgement message to the CU-CP **808**. The CU-CP **808** becomes aware from RRC Reconfigure Acknowledgement message that the

UE **802**, uniquely identified to the CU-CP **808** by the serving DU **804**, acknowledges completion of a successful serving cell change.

Also in response to receiving a layer 3 RRC measurement configuration, the UE **802** transmits **812** an RRC measurement report to the CU-CP **808**, in accordance with 3GPP standards. In accordance with 3GPP standards, the RRC measurement report can include layer 3 (L3) measurements that can be analyzed by the CU-CP **808** in making resource control decisions, which can include deciding to prepare at least one target DU cell for LTM so the at least one target cell from the target DU (also referred to herein as “neighbour DU”) **806** is ready to serve the UE **802** instead of the serving DU **804** for at least one service.

In response to deciding to prepare at least one target cell for LTM, the CU-CP **808** prepares **706** at least one target cell for LTM. As shown in FIG. **8**, in this illustrated embodiment, each of the at least one target cells is an inter-DU target cell, e.g., is part of a different DU than the serving DU **804** where the same CU (e.g., the CU that includes the CU-CP **808**) serves each DU **804**, **806** is part of a different DU as the serving DU **804**, but the same base station as the serving DU **804**. Also in this illustrated embodiment, the at least one target cell includes only the target DU **806** since there are only two gNB-DUs, but as mentioned above, the base station can include more than two target cells. Currently, in accordance with 3GPP standards, a maximum of eight LTM target cells can be prepared for a given UE.

Preparing **706** the at least one target cell for LTM can include notifying the at least one target DU **806** that the at least one target DU **806** may be later notified to begin providing service to the UE **802** for at least one service. The target DU **806** can thus reserve necessary resources for the UE **802**. As shown in FIG. **8**, the preparation **706** of the at least one target cell, which in this illustrated implementation is only the target DU **806**, can include the CU-CP **808** transmitting **814** a UE Context Setup Request message to the target DU **806**, in accordance with 3GPP standards, using an F1 communication interface.

In response to receiving the UE Context Setup Request message from the CU-CP **808**, the target DU **806** prepares each of the one or more target cells for LTM, e.g., reserves necessary resources for the UE **802**, and notifies the CU-CP **808** that the preparation **828** has been completed. In this illustrated implementation the at least one target cell includes only the target DU **806** preparing one target cell. In an inter-DU LTM scenario, one or more of the at least one LTM-prepared target cells belong to a different DU than the serving DU **806**. For example, with reference to the system of FIG. **6b**, the serving DU can be a small cell **626** of macro1 DU **628a**, and one or more of the target cells can be one or more small cells **626** of macro2 DU **628b** and/or macro3 DU **628c**. As shown in FIG. **8**, the notification to the CU-CP **808** can include the target DU **806** transmitting **816** a UE Context Setup Response message to the CU-CP **808**, in accordance with 3GPP standards, using an F1 communication interface. As also shown in FIG. **8**, the UE Context Setup Response message includes consolidated cell group configuration information for the one or more target cells prepared at the target DU **806**. The consolidated cell group configuration information includes the PCI (or other unique identifier) for each of the prepared one or more target cells.

In response to being notified that at least one target cell has been prepared by the target DU **806** for LTM, the CU-CP **808** maps **708**, **818** each prepared at least one target cell to an index. The mapping **708** includes, as shown in FIG. **8**, the CU-CP **808** creating **818** indexes for all target cells config-

ured as candidate L1/L2 centric inter-cell mobility targets for the UE **802**. A maximum number of the target cells prepared for the UE **802** is eight, per current 3GPP standards. The CU-CP **808** therefore maps **708** up to eight target cells under current 3GPP standards.

The mapping **708**, **818** of the at least one target cells includes correlating each of the at least one target cells to an index. As mentioned above, the CU-CP **808** is informed by the target DU **806** of the PCI (or other unique identifier) for each of the prepared one or more target cells. The CU-CP **808** can thus map each of the one or more target cells by PCI (or other unique identifier) to the index such that each of the prepared one or more target cells can be uniquely identified by the index. The CU-CP **808** stores the mapping between target cell PCI (or other unique identifier) and index of the target cell for future use, such as by storing the mapping as a table in a memory.

In some implementations, the index can be a numerical value such that each of the one or more target cells is associated by PCI (with or other unique identifier) a unique numerical value. Since the maximum number of the target cells prepared for the UE **802** is eight, per current 3GPP standards, eight different numerical values can be used for the index e.g., integers 1 to 8, integers 0 to 7, even integers 2 to 16, or other numerical values. For example, in an implementation in which three target cells have been prepared for LTM, the mapping **708**, **818** can include correlating a first PCI of a first one of the target to an index of 1, correlating a second PCI of a second one of the target to an index of 2, and correlating a third PCI of a third one of the target to an index of 3.

In some implementations, the index can be an alphabetical value such that each of the one or more target cells is associated by PCI (with or other unique identifier) a unique alphabetical value. Since the maximum number of the target cells prepared for the UE **802** is eight, per current 3GPP standards, eight different alphabetical values can be used for the index e.g., letters A to H, terms one to eight, or other alphabetical values. For example, in an implementation in which three target cells have been prepared for LTM, the mapping **708**, **818** can include correlating a first PCI of a first one of the target to an index of A, correlating a second PCI of a second one of the target to an index of B, and correlating a third PCI of a third one of the target to an index of C.

In some implementations, the index can be an alphanumeric value such that each of the one or more target cells is associated by PCI (with or other unique identifier) a unique alphanumeric value. Since the maximum number of the target cells prepared for the UE **802** is eight, per current 3GPP standards, eight different alphanumeric values can be used for the index e.g., terms cell 1 to cell8, terms C1 to C8, or other alphanumeric values. For example, in an implementation in which three target cells have been prepared for LTM, the mapping **708**, **818** can include correlating a first PCI of a first one of the target to an index of cell1, correlating a second PCI of a second one of the target to an index of cell2, and correlating a third PCI of a third one of the target to an index of cell3.

The CU-CP **808** can create mappings for each of a plurality of UEs, where the UE **802** is one of the plurality of UEs. Multiple mappings may thus be stored at a same time. Since each UE of the plurality of UEs will have its own mapping created as different UEs can have different LTM target cell candidates, the same target cell may be associated with a different index in different mappings. e.g., associated with an index of A in a first mapping for a first UE and associated with an index of D in a second mapping for a

second UE, associated with an index of 3 in a first mapping for a first UE and associated with an index of 5 in a second mapping for a second UE, etc. Different indexes in different mappings for the same PCI will not cause any confusion at the CU-CP 808, the serving DU 804, the target DU 806, or any of the plurality of UEs because each mapping is associated with and used in relation to a particular UE.

After the mapping 708, 818 has been performed, the CU-CP 808 notifies 710 the serving DU 804 of the mapping 708, 818 and the LTM-prepared at least one target cell. The serving DU 804 is thus made aware of each of the prepared one or more target cells by PCI (or other unique identifier) and by index. As shown in FIG. 8, the notification 710 to the serving DU 804 can include the CU-CP 808 transmitting 820 a UE Context Modification Request message to the serving DU 804 using an F1 communication interface. As also shown in FIG. 8, the UE Context Modification Request message can include, e.g., as an information element (IE), the mapping that correlates the PCI (or other unique identifier) each of the prepared one or more targets cells with an index. The serving DU 804 thus securely receives identification information of each of the one or more target cells configured as candidate L1/L2 centric inter-cell mobility targets for the UE 802.

In response to being notified 710 of the at least one LTM prepared target DU cell, the serving cell 804 stores the received information regarding the at least one LTM target cell, e.g., stores the mapping. Also in response to being notified 710 of the at least one target DU, the serving cell 804 transmits 822 a UE Context Modification Response message to the CU-CP 808 using an F1 communication interface. As shown in FIG. 8, the UE Context Modification Response message can include consolidated cell group configuration information for each the one or more target cells identified to the UE 802 by the CU-CP 808. The UE Context Modification Request message and the UE Context Modification Response message are each defined by 3GPP. The serving DU 804 can thus receive information regarding the at least one target cell from the CU-CP 808, and can acknowledge the receipt to the CU-CP 808, using messages already transmitted for HO in accordance with 3GPP standards.

In response to receiving the UE Context Modification Response message, the CU-CP 808 transmits 824 an RRC reconfiguration message to the UE 802, in accordance with 3GPP standards. As also shown in FIG. 8, the RRC reconfiguration message includes LTM target cell configuration information, e.g., as provided to the CU-CP 808 from the target DU 806 in the transmitted 830 UE Context Setup Response message, and includes the index created by the CU-CP 808 for the one or more target cells configured as candidate L1/L2 centric inter-cell mobility targets for the UE 802. The UE 802 is thus notified 712 of the LTM target cell configuration information and the index by the CU-CP 808. Further, the UE 802 securely receives identification information of each of the one or more target cells configured as candidate L1/L2 centric inter-cell mobility targets for the UE 802 since the index is transmitted 824 to the UE 802 in a secure message.

In response to receiving the target cell configurations in the RRC reconfiguration message, the UE 802 transmits 826 an L1 measurement report to the serving DU 804, in accordance with 3GPP standards. The L1 measurement report provides the UE measured radio condition information of the configured one or more target cells to the serving DU 804.

In response to receiving the L1 measurement report transmitted 826 from the UE 802, the serving cell 804 selects 714, 828 a target cell, from among the one or more LTM-prepared target cells identified to the serving DU 804. In this illustrated embodiment there is only one target cell (the target DU 806) identified to the serving DU 804 by the CU-CP 808 as an LTM-prepared target cell, so the serving cell's selection 714, 828 is straightforward, with the serving DU 804 selecting 714, 828 the target DU 806. If there are multiple LTM-prepared target cells which satisfy the handover criteria at the serving DU 804, the serving DU 804 is configured to select one of the target cells for handover.

In implementations in which there are a plurality of target cells identified to the serving DU 804 by the CU-CP 808, the serving cell's target cell selection 714, 828 can include determining which one or more of the plurality of target cells has a radio quality above a predetermined threshold radio quality, as shown in FIG. 8. The predetermined threshold radio quality is defined by the UE's radio condition that the serving DU 804 received from the UE 802 in the L1 measurement report. The serving DU 804 can thus take into consideration particular needs of the particular UE 802 involved in the HO when selecting 710, 828 a target cell for the HO. Additionally, the L1 measurement report transmitted 826 by the UE 802 to the serving DU 804 reports L1 measurements, which can include reference signal received power (RSRP) as defined by 3GPP, for each of the plurality of target cells, whose identities are known by the UE 802 as having been provided to the UE 802 by the CU-CP 808 in the RRC reconfiguration message. The serving DU 804 can therefore analyze the L1 measurement report received from the UE 802 to determine which one or more of the plurality of target cells has a radio quality above a predetermined threshold radio quality.

If only one of the plurality of target cells satisfies the UE's radio condition, e.g., only one of the target cells' radio qualities is above the predetermined threshold radio quality, then the serving cell 804 selects 714, 828 that target cell. If more than one of the plurality of target cells satisfies the UE's radio condition, e.g., the target cells' radio qualities are each above the predetermined threshold radio quality, then any one of these target cells would be able to serve the UE's needs at random or according to another desired criteria.

Having selected 710, 828 the target cell (e.g., the target DU 806 in the illustrated implementation of FIG. 8), the serving DU 804 triggers 716 serving cell change to the selected 714, 828 target cell. The serving DU 804 identifies the selected 714, 828 target cell to the UE 802 by the index, e.g., number, alphabetical value, etc., associated with the selected 714, 828 target cell per the mapping performed by the CU-CP 808. The serving DU 804 therefore does not need to transmit the PCI (or other unique identifier) of the selected 714, 828 target cell at all to the UE 802, in an unsecure message or otherwise. Security may therefore be improved because even in the event of a security breach in which the index of the selected 714, 828 target cell becomes known to an unauthorized party, the index, as a value that does not uniquely identify the target cell or the UE 802 except to an authorized party that has the mapping, cannot allow the user's path to be tracked or the user's location to be traced. As shown in FIG. 8, triggering 716 the serving cell change can include the serving DU 804 transmitting 844 a MAC CE to the UE 802 that includes a serving cell change command and identifies the selected 714, 828 target cell to the UE 802. Thus, even though the MAC CE is an unsecure message, the selected 714, 828 target cell can be identified to the UE 802 without any security compromise.

Also after having selected **710**, **828** the target cell, the serving DU **804** transmits **832** a serving cell change (SCC) notification to the CU-CP **808**, via an F1 communications interface, that identifies the target DU **806**, e.g., by PCI (or other unique identifier), as the new; current serving cell for the UE **802** for at least one service.

The UE's receipt of the MAC CE indicates to the UE **802** that LTM serving cell change (SCC) has to be performed to the identified target cell, e.g., the target DU **806** in the illustrated implementation of FIG. **8**, for the UE **802**. Thus, in response to receiving the MAC CE from the serving cell **804**, the UE **802** begins **718** HO to the target cell. The HO can be performed in accordance with 3GPP standards. The UE determines the identity of the target cell for HO using the index received from the serving DU **804**. The UE **802** received the mapping from the CU-CP **808** and can therefore look up in the index received from the serving DU **804** in the mapping to determine the target cell that corresponds to the index, thereby identifying the target cell for HO. As shown in FIG. **8a**, the UE **802** beginning **718** HO to the target cell can include the UE **802** accessing **834** the target cell in a RACH procedure for a RACH-based HO, which can be performed in accordance with 3GPP standards. As shown in FIG. **8**, the UE's access **834** of the target cell can include the UE **802** transmitting a preamble to the target DU **806**.

Also in response to receiving the MAC CE from the serving cell **804**, the UE **802** transmits **840** an RRC reconfiguration acknowledgment message to the CU-CP **808**. The CU-CP **808** therefore receives acknowledgment from both the UE **802**, via the RRC reconfiguration acknowledgment message indicating a successful RRC reconfiguration at the UE **802**, and the serving DU **804**, via the serving cell change notification, that the target DU **806** is now serving the UE **802** for at least one service that has been handed over from the serving DU **804**.

In response to receiving the SCC notification from the serving DU **804**, the CU-CP **808** notifies **720** the target DU **806** of the mapping created by the CU-CP **808** for the one or more target cells configured as candidate L1/L2 centric inter-cell mobility targets for the UE **802**. The target DU **806** thus securely receives identification information of each of the one or more target cells configured as candidate L1/L2 centric inter-cell mobility targets for the UE **802**. The target DU **806** can use the mapping in the event that HO for the UE **802** occurs from the target DU **806**, similar to that discussed above regarding the serving DU **804** handing over to the target DU **806**. As shown in FIG. **8**, the notification **820** can include the CU-CP **808** transmitting **836** a UE Context Modification Request message to the target DU **806** using an F1 communication interface. As shown in FIG. **8**, the UE **802** can include, e.g., as an information element (IE), the mapping. In response to receiving the UE Context Modification Request message from the CU-CP **808**, the target DU **806** transmits **838** a UE Context Modification Response message to the CU-CP **808** using an F1 communication interface.

After the CU-CP **808** has created **708**, **818** the mapping for the one or more target cells configured as candidate L1/L2 centric inter-cell mobility targets for the UE **802**, the CU-CP **808** is configured to update the mapping. Updating the mapping allows for the mapping to be a dynamic, real time reflection of target cells that are LTM candidates for HO for the UE **802**. The CU-CP **808** is configured to provide the updated mapping to the UE **802** in a secure message similar to that discussed above regarding the CU-CP **808** transmitting **824** the mapping to the UE **802**. The CU-CP **808** is also configured to provide the updated mapping to the

serving DU **804** (which may be the target DU **806** if the mapping is updated after HO to the selected **714**, **828** target cell has already occurred) in a secure message similar to that discussed above regarding the CU-CP **808** transmitting **820** the mapping to the serving DU **804** and transmitting **836** the mapping to the target DU **806**. The UE **802** and the serving DU **804** (which may be the target DU **806**) can thus have current mapping for the UE **802** in the event that HO for the UE **802** occurs after the mapping has been updated.

The CU-CP **808** is configured to update the mapping for the one or more target cells configured as candidate L1/L2 centric inter-cell mobility targets for the UE **802** in response to occurrence of a trigger event. One example of the trigger event is one of the target cells being removed for the UE **802**, such as if the target cell goes offline or gets removed from the wireless communications system that includes the CU-CP **808**. The CU-CP **808** is configured to delete the index for the removed target cell from the mapping, e.g., deletes the entry for that target cell from the table that correlates target cells to indexes.

Another example of the trigger event is one of the target cells being replaced for the UE **802**, such as if a target cell not previously a candidate becomes a better candidate than one of the target cells currently identified as a candidate, e.g., because of changed requirements of the UE **802**. The CU-CP **808** is configured to delete the index for the target cell being replaced from the mapping, e.g., deletes the entry for that target cell from the table that correlates target cells to indexes, and to associate that index instead with the target cell being added as a candidate.

When the mapping for the one or more target cells configured as candidate L1/L2 centric inter-cell mobility targets for the UE **802** is updated, the index may become non-sequential. There is no need to re-assign the PCIs to indexes to make them sequential since the indexes will still be valid. For example, if integers of 1 to 8 have been used for eight target cells and the target cell associated with an index of 2 is removed the index will no longer be sequential for the remaining target cells that are associated with indexes of 1, 3, 4, 5, 6, 7, and 8. Should another target cell be added to this mapping the target cell that is added can be associated with the now unused index of 2.

FIG. **9** illustrates an exemplary architecture of the gNB-CU-CP **808** of FIG. **8**. As shown in FIG. **9**, the gNB-CU-CP **808** includes a memory **900**, a processor **902**, a communicator **904**, and a security management controller **906**. The communicator **1106** is configured to communicate internally between internal hardware components of the CU-CP **808** and with external devices via one or more networks. The communicator **904** can include an electronic circuit specific to a standard that enables wired or wireless communication. The security management controller **906** is configured to perform the mapping discussed above, and the memory **900** is configured to store the mapping therein. Multiple mappings can be stored in the memory **900**, one mapping per UE such that the CU-CP **808** maintains mappings for multiple UEs. Although FIG. **9** shows hardware components of the gNB-CU-CP **808**, other implementations of the CU-CP **808** are possible. For example, the gNB-CU-CP **808** may include less or a greater number of components.

In some implementations, the current subject matter can be configured to be implemented in a system **1000**, as shown in FIG. **10**. The system **1000** can include one or more of a processor **1010**, a memory **1020**, a storage device **1030**, and an input/output device **1040**. Each of the components **1010**, **1020**, **1030** and **1040** can be interconnected using a system bus **1050**. The processor **1010** can be configured to process

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instructions for execution within the system 600. In some implementations, the processor 1010 can be a single-threaded processor. In alternate implementations, the processor 1010 can be a multi-threaded processor. The processor 1010 can be further configured to process instructions stored in the memory 1020 or on the storage device 1030, including receiving or sending information through the input/output device 1040. The memory 1020 can store information within the system 1000. In some implementations, the memory 1020 can be a computer-readable medium. In alternate implementations, the memory 1020 can be a volatile memory unit. In yet some implementations, the memory 1020 can be a non-volatile memory unit. The storage device 1030 can be capable of providing mass storage for the system 1000. In some implementations, the storage device 1030 can be a computer-readable medium. In alternate implementations, the storage device 1030 can be a floppy disk device, a hard disk device, an optical disk device, a tape device, non-volatile solid state memory, or any other type of storage device. The input/output device 1040 can be configured to provide input/output operations for the system 1000. In some implementations, the input/output device 1040 can include a keyboard and/or pointing device. In alternate implementations, the input/output device 1040 can include a display unit for displaying graphical user interfaces.

FIG. 11 illustrates an exemplary method 1100 for executing secure layer 1/layer 2 (L1/L2) centric inter-cell mobility, according to some implementations of the current subject matter. The method 1100 may be performed, for example, using implementations shown in and described with respect to FIGS. 6a-9.

The method 1100 includes mapping 1102 a PCI of at least one LTM target cell of at least one target DU of a base station that is a candidate for HO for a UE to an index, and securely transmitting 1104 the mapping from a CU-CP of the base station to the UE and to a serving DU of the base station that is currently serving the UE for at least one service.

In some implementations, the current subject matter can include one or more of the following optional features.

In some implementations, the index for each of the at least one LTM targets cells can be unique in the mapping such that each index is uniquely associated with one of the at least one LTM target cells by PCI.

In some implementations, the method can also include securely transmitting the mapping from the CU-CP to a target DU including a one of the at least one LTM targets cells selected for the HO.

In some implementations, the mapping can be transmitted to the UE in a radio resource control (RRC) reconfiguration message, and the mapping can be transmitted to the serving DU in an F1 UE context modification request message.

In some implementations, the method can also include updating the mapping in response to at least one of: one of the at least one LTM targets cells being removed for the UE, and one of the at least one LTM targets cells being replaced for the UE, and the method can also include securely transmitting the updated mapping from the CU-CP to the UE and the serving DU.

In some implementations, the method can also include mapping a PCI of at least one LTM target cell of the at least one target DU that is a candidate for HO for a second UE to an index such that a second mapping is created, and securely transmitting the second mapping from the CU-CP to the second UE.

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In some implementations, the serving DU can utilize the index received from the CU-CP when transmitting an LTM HO command to the UE.

In some implementations, each index can be one of numerical, alphabetical, and alphanumeric.

In some implementations, the method can also include storing the mapping in the at least one non-transitory storage media.

In some implementations, the base station can include a Next Generation Radio Access network (NG-RAN) node. Further, the NG-RAN node can include a gNodeB or an ng-eNodeB.

In some implementations, the base station can include the at least one processor and the at least one non-transitory storage media. Further, the CU-CP can include the at least one processor and the at least one non-transitory storage media.

The systems and methods disclosed herein can be embodied in various forms including, for example, a data processor, such as a computer that also includes a database, digital electronic circuitry, firmware, software, or in combinations of them. Moreover, the above-noted features and other aspects and principles of the present disclosed implementations can be implemented in various environments. Such environments and related applications can be specially constructed for performing the various processes and operations according to the disclosed implementations or they can include a general-purpose computer or computing platform selectively activated or reconfigured by code to provide the necessary functionality. The processes disclosed herein are not inherently related to any particular computer, network, architecture, environment, or other apparatus, and can be implemented by a suitable combination of hardware, software, and/or firmware. For example, various general-purpose machines can be used with programs written in accordance with teachings of the disclosed implementations, or it can be more convenient to construct a specialized apparatus or system to perform the required methods and techniques.

The systems and methods disclosed herein can be implemented as a computer program product, i.e., a computer program tangibly embodied in an information carrier, e.g., in a machine readable storage device or in a propagated signal, for execution by, or to control the operation of, data processing apparatus, e.g., a programmable processor, a computer, or multiple computers. A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program can be deployed to be executed on one computer or on multiple computers at one site or distributed across multiple sites and interconnected by a communication network.

As used herein, the term "user" can refer to any entity including a person or a computer.

Although ordinal numbers such as first, second, and the like can, in some situations, relate to an order: as used in this document ordinal numbers do not necessarily imply an order. For example, ordinal numbers can be merely used to distinguish one item from another. For example, to distinguish a first event from a second event, but need not imply any chronological ordering or a fixed reference system (such that a first event in one paragraph of the description can be different from a first event in another paragraph of the description).

The foregoing description is intended to illustrate but not to limit the scope of the invention, which is defined by the

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scope of the appended claims. Other implementations are within the scope of the following claims.

These computer programs, which can also be referred to as programs, software, software applications, applications, components, or code, include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the term “machine-readable medium” refers to any computer program product, apparatus and/or device, such as for example magnetic discs, optical disks, memory, and Programmable Logic Devices (PLDs), used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor. The machine-readable medium can store such machine instructions non-transitorily, such as for example as would a non-transient solid state memory or a magnetic hard drive or any equivalent storage medium. The machine-readable medium can alternatively or additionally store such machine instructions in a transient manner, such as for example as would a processor cache or other random access memory associated with one or more physical processor cores.

To provide for interaction with a user, the subject matter described herein can be implemented on a computer having a display device, such as for example a cathode ray tube (CRT) or a liquid crystal display (LCD) monitor for displaying information to the user and a keyboard and a pointing device, such as for example a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well. For example, feedback provided to the user can be any form of sensory feedback, such as for example visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including, but not limited to, acoustic, speech, or tactile input.

The subject matter described herein can be implemented in a computing system that includes a back-end component, such as for example one or more data servers, or that includes a middleware component, such as for example one or more application servers, or that includes a front-end component, such as for example one or more client computers having a graphical user interface or a Web browser through which a user can interact with an implementation of the subject matter described herein, or any combination of such back-end, middleware, or front-end components. The components of the system can be interconnected by any form or medium of digital data communication, such as for example a communication network. Examples of communication networks include, but are not limited to, a local area network (“LAN”), a wide area network (“WAN”), and the Internet.

The computing system can include clients and servers. A client and server are generally, but not exclusively, remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

The implementations set forth in the foregoing description do not represent all implementations consistent with the subject matter described herein. Instead, they are merely some examples consistent with aspects related to the described subject matter. Although a few variations have

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been described in detail above, other modifications or additions are possible. In particular, further features and/or variations can be provided in addition to those set forth herein. For example, the implementations described above can be directed to various combinations and sub-combinations of the disclosed features and/or combinations and sub-combinations of several further features disclosed above. In addition, the logic flows depicted in the accompanying figures and/or described herein do not necessarily require the particular order shown, or sequential order, to achieve desirable results. Other implementations can be within the scope of the following claims.

What is claimed:

1. An apparatus, comprising:

at least one processor; and

at least one non-transitory storage media storing instructions that, when executed by the at least one processor, cause the at least one processor to perform operations comprising:

mapping a physical cell identifier (PCI) of at least one layer 1/layer 2 triggered mobility (LTM) target cell of at least one target distributed unit (DU) of a base station that is a candidate for handover (HO) for a user equipment (UE) to an index; and

securely transmitting the mapping from a centralized unit control plane (CU-CP) of the base station to the UE and to a serving DU of the base station that is currently serving the UE for at least one service.

2. The apparatus of claim 1, wherein the index for each of the at least one LTM targets cells is unique in the mapping such that each index is uniquely associated with one of the at least one LTM target cells by PCI.

3. The apparatus of claim 1, wherein the operations further comprise securely transmitting the mapping from the CU-CP to a target DU including a one of the at least one LTM targets cells selected for the HO.

4. The apparatus of claim 1, wherein the mapping is transmitted to the UE in a radio resource control (RRC) reconfiguration message; and

the mapping is transmitted to the serving DU in an F1 UE context modification request message.

5. The apparatus of claim 1, wherein the operations further comprise:

updating the mapping in response to at least one of:

one of the at least one LTM targets cells being removed for the UE, and

one of the at least one LTM targets cells being replaced for the UE; and securely transmitting the updated mapping from the CU-CP to the UE and the serving DU.

6. The apparatus of claim 1, wherein the operations further comprise:

mapping a PCI of at least one LTM target cell of the at least one target DU that is a candidate for HO for a second UE to an index such that a second mapping is created; and

securely transmitting the second mapping from the CU-CP to the second UE.

7. The apparatus of claim 1, wherein the serving DU utilizes the index received from the CU-CP when transmitting an LTM HO command to the UE.

8. The apparatus of claim 1, wherein each index is one of numerical, alphabetical, and alphanumeric.

9. The apparatus of claim 1, wherein the operations further comprise storing the mapping in the at least one non-transitory storage media.

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10. The apparatus of claim 1, wherein the base station includes a Next Generation Radio Access network (NG-RAN) node.

11. The apparatus of claim 10, wherein the NG-RAN node includes a gNodeB or a ng-eNodeB.

12. The apparatus of claim 1, wherein the base station includes the at least one processor and the at least one non-transitory storage media.

13. The apparatus of claim 12, wherein the CU-CP includes the at least one processor and the at least one non-transitory storage media.

14. At least one non-transitory storage media storing instructions that, when executed by at least one processor, cause the at least one processor to perform operations comprising:

mapping a physical cell identifier (PCI) of at least one layer 1/layer 2 triggered mobility (LTM) target cell of at least one target distributed unit (DU) of a base station that is a candidate for handover (HO) for a user equipment (UE) to an index; and

securely transmitting the mapping from a centralized unit control plane (CU-CP) of the base station to the UE and to a serving DU of the base station that is currently serving the UE for at least one service.

15. The non-transitory storage media of claim 14, wherein the index for each of the at least one LTM targets cells is

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unique in the mapping such that each index is uniquely associated with one of the at least one LTM target cells by PCI.

16. The non-transitory storage media of claim 14, wherein the operations further comprise securely transmitting the mapping from the CU-CP to a target DU including a one of the at least one LTM targets cells selected for the HO.

17. A computer-implemented method, comprising:

mapping a physical cell identifier (PCI) of at least one layer 1/layer 2 triggered mobility (LTM) target cell of at least one target distributed unit (DU) of a base station that is a candidate for handover (HO) for a user equipment (UE) to an index; and

securely transmitting the mapping from a centralized unit control plane (CU-CP) of the base station to the UE and to a serving DU of the base station that is currently serving the UE for at least one service.

18. The method of claim 17, wherein the index for each of the at least one LTM targets cells is unique in the mapping such that each index is uniquely associated with one of the at least one LTM target cells by PCI.

19. The method of claim 17, further comprising securely transmitting the mapping from the CU-CP to a target DU including a one of the at least one LTM targets cells selected for the HO.

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